AK09973D

3D Magnetic Smart Switch Sensor

1. General Description

AK09973D is a 3D magnetic smart switch sensor IC with high sensitivity and wide measurement range utilizing our latest Hall sensor technology.
Our ultra-small package of AK09973D incorporates magnetic sensors, chopper stabilized signal amplifier chain, and all necessary interface logic for detecting weak to strong magnetic fields in the X-axis, Y-axis and Z-axis independently. From its compact foot print, thin package, and extremely low power consumption, it is suitable for a smartphone and wearable application.

2. Features

- Functions:
  - 16-bit data out for each 3-axis magnetic component
  - Built-in A to D Converter for magnetometer data output
  - Sensor measurement range and sensitivity
    - High sensitivity setting
      - Sensitivity: 1.1 µT/LSB (typ.)
      - Measurement range: ±36 mT
    - Wide range setting
      - Sensitivity: 3.1 µT/LSB (typ.)
      - Measurement range: X and Y-axis → ±34.9 mT, Z-axis → ±101.5 mT
  - Serial interface
    - I²C bus interface
      Standard mode, Fast mode and Fast mode plus
    - Two selectable slave addresses
  - Operation mode
    - Power-down, Single measurement, Continuous measurement, Self-test
  - 3-axis programmable switch function
  - Output pin for event notification
    - OD-INT pin
  - DRDY function for measurement data ready
  - Magnetic sensor overflow monitor function
  - Built-in power on reset function
  - Built-in oscillator for internal clock source
  - Selectable sensor drive
    - Low power drive / Low noise drive
- Operating temperature:
  - -30˚C to +85˚C
- Operating supply voltage:
  - 1.65 V to 1.95 V
- Current consumption (VDD = 1.8 V, +25˚C):
  - Power-down: 1.0 µA (typ.)
  - Measurement:
    - Average current consumption at 10 Hz / 100 Hz repetition rate
      - Low power drive: 3.5 µA(typ.) at 10 Hz, 22 µA(typ.) at 100 Hz
      - Low noise drive: 11 µA(typ.) at 10 Hz, 101 µA(typ.) at 100 Hz
- Package
  - AK09973D 5-pin WL-CSP (BGA) package: 1.18 mm × 0.78 mm × 0.55 mm

09973D三维磁智能开关传感器

1. 概述

AK09973D是一款采用我们最新霍尔传感器技术的三维磁智能开关传感器集成电路，具有高灵敏度和宽测量范围。

AK09973D 采用超小型封装，集成了磁传感器、斩波器稳定信号放大链以及所有必要的接口逻辑电路，可独立检测 X 轴、Y 轴和 Z 轴上的弱磁场至强磁场。其紧凑的封装尺寸、纤薄的厚度以及极低的功耗，使其非常适合智能手机和可穿戴应用。

2. 特性

◆ 功能：
➢ 16位三轴磁分量数据输出➢ 内置用于磁力计数据输出的模数转换器➢ 传感器测量范围和灵敏度✧ 高灵敏度设置

● 灵敏度：1.1 微特斯拉/最低有效位（典型值）
● 测量范围：±36 毫特斯拉✧ 宽范围设置

● 灵敏度：3.1 微特斯拉/最低有效位（典型值）
● 测量范围：X轴和Y轴 → ±34.9 毫特斯拉，Z轴 → ±101.5 毫特斯拉➢ 串行接口✧ I²C总线接口 标准模式、快速模式和增强快速模式✧ 两个可选从机地址
➢ 操作模式✧ 掉电模式、单次测量、连续测量、自检➢ 3轴可编程开关功能
➢ 用于事件通知的输出引脚✧ OD-INT引脚➢ 用于测量数据就绪的数据就绪功能➢ 磁传感器溢出监控功能➢ 内置上电复位功能➢ 用于内部时钟源的内置振荡器➢ 可选传感器驱动✧ 低功耗驱动 / 低噪声驱动

◆ 工作温度：➢ -30℃ 至 +85˚C
◆ 工作电源电压：➢ 1.65 V 至 1.95 V
◆ 电流消耗（VDD = 1.8 V，+25℃）：➢ 掉电模式：1.0 微安（典型值）
➢ 测量：✧ 在 10 赫兹 / 100 赫兹 重复频率下的平均电流消耗

● 低功耗驱动：10赫兹下为3.5微安（典型值），100赫兹下为22微安（典型值）
● 低噪声驱动：10赫兹下为11微安（典型值），100赫兹下为101微安（典型值）
◆ 封装➢ AK09973D 5引脚WL-CSP (BGA) 封装：1.18 毫米 × 0.78 毫米 × 0.55 毫米

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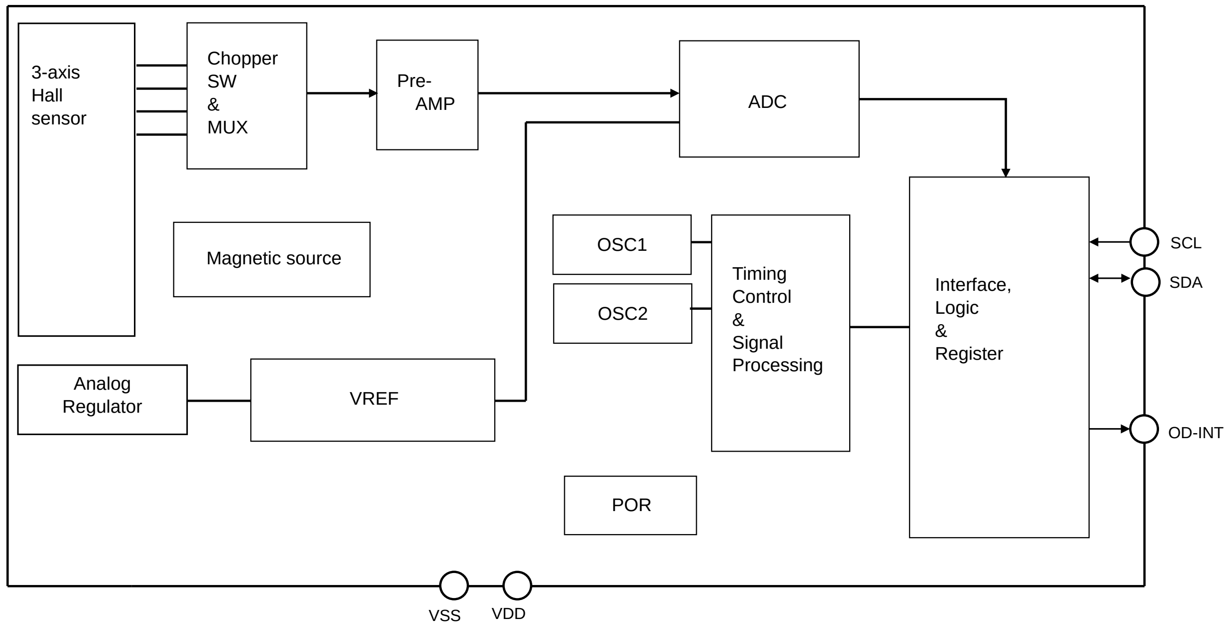
4. Block Diagram and Functions

4.1. Block Diagram

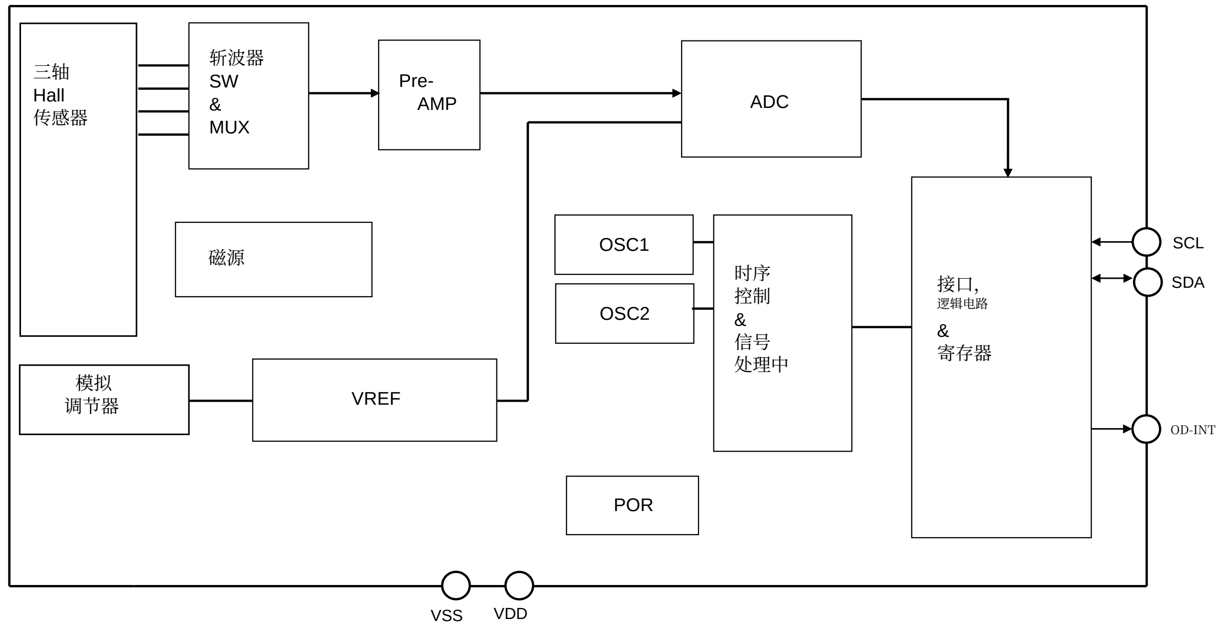


4.2. Functions

Block	Function
3-axis Hall sensor	Monolithic Hall elements.
Chopper SW & MUX	Multiplexer for selecting Hall elements.
Analog Regulator	Internal power supply.
Pre-AMP	Differential amplifier used to amplify the magnetic sensor signal.
ADC	Convert analog output to digital output.
OSC1	Generates an operating clock for sensor measurement.
OSC2	Generates an operating periodic clock for sequencer.
POR	Power On Reset circuit. Generates reset signal on rising edge of VDD.
VREF	Generates temperature independent reference voltage.
Interface Logic & Register	Exchanges data with an external CPU. OD-INT pin indicates some magnetic event (selectable). I^2C bus interface using two pins (SCL and SDA). Standard mode, Fast mode and Fast mode plus are supported.
Timing Control & Signal Processing	Generates a timing signal required for internal operation. Magnetic sensitivity adjustment and switch calculation for switch function.
Magnetic source	Generates magnetic field for Self-test of magnetic sensor.

4. 框图与功能

4.1. 框图



4.2. 功能

模块	功能
三轴霍尔传感器	单片霍尔元件。
斩波开关与多路复用器	用于选择霍尔元件的多路复用器。
模拟稳压器	内部电源。
前置放大器	差分放大器用于放大磁传感器信号。
ADC	将模拟输出转换为数字输出。
OSC1	为传感器测量生成工作时钟。
OSC2	为序列发生器生成运行周期时钟。
POR	上电复位电路。在VDD上升沿产生复位信号。
VREF	生成温度无关基准电压。
接口逻辑 & 寄存器	与外部CPU交换数据。 OD-INT引脚指示某些磁性事件（可选）。 使用两个引脚（SCL和SDA）的I^2C总线接口。支持标准模式、快速模式和增强快速模式。
时序控制 & 信号处理	生成内部操作所需的时序信号。 用于开关功能的磁灵敏度调整和开关计算。
磁源	为磁传感器的自检生成磁场。

5. Pin Configurations and Functions

AK09973D has two pin connections shown in the tables below. The slave address can be switched by changing the pin connection.

Connection 1: Slave address = 10h

Pin No.	Pin name	Function name	I/O	Type	Function
B1	IF1	OD-INT	O	Digital	Open-drain interrupt pin “L” active. Refer to section 10.7. Connect this pin to VSS when not using OD-INT.
B3	IF2	SDA	I/O	Digital	Control data input/output pin Input: Schmitt trigger, Output: Open-drain
A1	VDD	VDD	-	Power	Positive power supply pin
A2	VSS	VSS	-	Ground	Ground pin
A3	SCL	SCL	I	Digital	Control clock input pin Input: Schmitt trigger

Connection 2: Slave address = 11h

Pin No.	Pin name	Function name	I/O	Type	Function
B1	IF1	SDA	I/O	Digital	Control data input/output pin Input: Schmitt trigger, Output: Open-drain
B3	IF2	OD-INT	O	Digital	Open-drain interrupt pin “L” active. Refer to section 10.7. Connect this pin to VSS when not using OD-INT.
A1	VDD	VDD	-	Power	Positive power supply pin
A2	VSS	VSS	-	Ground	Ground pin
A3	SCL	SCL	I	Digital	Control clock input pin Input: Schmitt trigger

5. 引脚配置与功能

AK09973D 具有如下表所示的两个引脚连接。其从机地址可通过更改引脚连接。

连接1：从机地址 = 10h

Pin No.	引脚名称	功能 name	I/O	Type	功能
B1	IF1	OD-INT	O	数字	开漏中断引脚 “L”有效。请参阅第10.7节。 不使用时，请将此引脚连接到VSS OD-INT。
B3	IF2	SDA	I/O	数字	控制数据输入/输出引脚 输入：施密特触发器，输出：开漏
A1	VDD	VDD	-	电源	正电源引脚
A2	VSS	VSS	-	接地	接地引脚
A3	SCL	SCL	I	数字	控制时钟输入引脚 输入：施密特触发器

连接2：从机地址 = 11h

Pin No.	引脚名称	功能 n name	输入/输出类型 功能		
B1	IF1	SDA	I/O	数字	控制数据输入/输出引脚 输入：施密特触发器，输出：开漏
B3	IF2	OD-INT	O	数字	开漏中断引脚 “L”有效。请参阅第10.7节。 不使用时请将此引脚连接至VSS OD-INT。
A1	VDD	VDD	-	电源	正电源引脚
A2	VSS	VSS	-	接地	接地引脚
A3	SCL	SCL	I	数字	控制时钟输入引脚 输入：施密特触发器

6. Absolute Maximum Ratings

Vss = 0 V

Parameter	Symbol	Min.	Max.	Unit
Power supply voltage	Vdd	-0.3	+2.5	V
Input voltage	VIN	-0.3	Vdd + 0.3	V
Input current	IIN	-10	+10	mA
Storage temperature	Tst	-40	+125	˚C

Note:
If the device is used in conditions exceeding these values, the device may be destroyed. Normal operations are not guaranteed in such exceeding conditions.

7. Recommended Operating Conditions

Vss = 0 V

Parameter	Symbol	Min.	Typ.	Max.	Unit
Operating temperature	Ta	-30	-	+85	˚C
Power supply voltage	Vdd	1.65	1.8	1.95	V
Input voltage	VIN	1.1	1.8	Vdd	V

8. Electrical Characteristics

The following conditions apply unless otherwise noted:
Vdd = 1.65 V to 1.95 V, VIN = 1.1 V to Vdd, Temperature range = -30˚C to +85˚C
Typical condition: Vdd = 1.8 V, Temperature = +25˚C

8.1. DC Characteristics

Parameter	Symbol	Pin	Condition	Min.	Typ.	Max.	Unit
High level input voltage* 1	VIH	SCL IF1 IF2	-	1.00	-	Vdd + 0.3	V
Low level input voltage* 1	VIL	SCL IF1 IF2	-	-0.3	-	0.42	V
Input current* 1	IIN	SCL IF1 IF2	VIN = Vss or Vdd	-10	-	+10	µA
Hysteresis input voltage* 2	VHS	SCL IF1 IF2	-	0.15	-	-	V
Low level output voltage	VOL	IF1 IF2	IOL* 3 ≤ +20mA	-	-	0.33	V
Current consumption* 4	IDD1	VDD	Power-down mode VIN = Vdd	-	1	3	µA
	IDD2		When magnetic sensor is driven	-	1.4	2.2	mA
	IDD3		When self-test is driven	-	5.6	-	mA

Notes:
* 1. As for IF1 and IF2 pins, the specification is applied when these pins are used for SDA function.
* 2. Schmitt trigger input (reference value for design).
* 3. IOL: Low level output current.
* 4. Without any resistance load.

6. 绝对最大额定值

VSS = 0 V

参数	符号	Min.	Max.	Unit
电源电压	Vdd	-0.3	+2.5	V
输入电压	VIN	-0.3	VDD + 0.3	V
输入电流	IIN	-10	+10	mA
存储温度	Tst	-40	+125	˚C

注意:
若设备在超出这些数值的条件下使用，设备可能被损坏。在此类超标条件下，无法保证正常工作。

7. 推荐工作条件

VSS = 0 V

参数	符号	Min.	Typ.	Max.	Unit
工作温度	Ta	-30	-	+85	˚C
电源电压	Vdd	1.65	1.8	1.95	V
输入电压	VIN	1.1	1.8	Vdd	V

8. 电气特性

除非另有说明，以下条件适用：Vdd = 1.65 V 至 1.95 V，VIN = 1.1 V 至 Vdd，温度范围 = -30℃ 至 +85℃典型条件：Vdd = 1.8 V，温度 = +25℃

8.1. 直流特性

参数	符号	Pin	条件	Min.	Typ.	Max.	Unit
高电平输入电压* 1	VIH	SCL IF1 IF2	-	1.00	-	VDD + 0.3	V
低电平输入电压*1	VIL	SCL IF1 IF2	-	-0.3	-	0.42	V
输入电流*1	IIN	SCL IF1 IF2	VIN = Vss 或 Vdd	-10	-	+10	µA
磁滞输入电压* 2	VHS	SCL IF1 IF2	-	0.15	-	-	V
低电平输出电压	VOL	IF1 IF2	IOL*3 ≤ +20毫安	-	-	0.33	V
电流消耗*4	IDD1	VDD	掉电模式 mode VIN = Vdd	-	1	3	µA
	IDD2		当磁传感器被驱动时	-	1.4	2.2	mA
	IDD3		当自检被驱动	-	5.6	-	mA

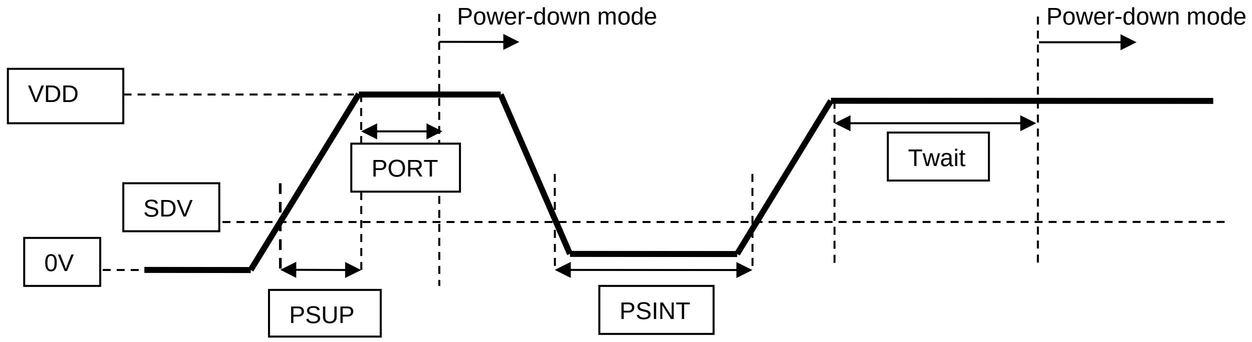
注：* 1. 对于IF1和IF2引脚，此规格适用于这些引脚用作SDA功能时。* 2. 施密特触发器输入（设计参考值）。* 3. IOL：低电平输出电流。* 4. 无任何电阻负载。

8.2. AC Characteristics

Parameter	Symbol	Pin	Condition	Min.	Typ.	Max.	Unit
Power supply rise time* 5	PSUP	VDD	Period of time that VDD changes from 0.2 V to Vdd.	-	-	50	ms
POR completion time* 5	PORT		Period of time after PSUP to Power-down mode* 6	-	-	100	µs
Power supply turn off voltage* 5	SDV		Turn off voltage to enable POR to restart* 6	-	-	0.2	V
Power supply turn on interval* 5	PSINT		Period of time that voltage lower than SDV needed to be kept to enable POR to restart	100	-	-	µs
Wait time before mode setting	Twait	-	-	100	-	-	µs

Notes:
* 5. Reference value for design.
* 6. When POR circuit detects the rise of VDD voltage, it resets internal circuits and initializes the registers. After reset, AK09973D transits to Power-down mode.

[Voltage waveform of VDD]

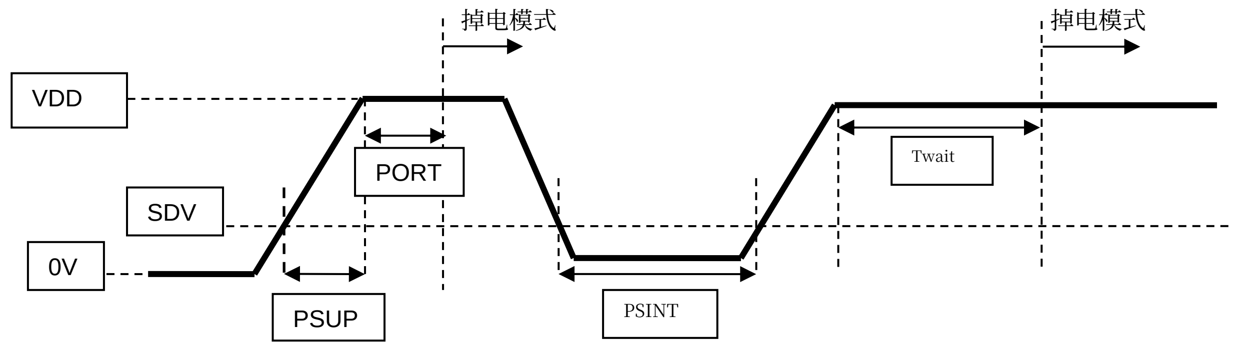


8.2. 交流特性

参数	Symbol	Pin	Condition	Min.	Typ.	最大值	单位
Power supply rise time* 5	PSUP	VDD	VDD 从 0.2 V 变化至 Vdd 的 从 0.2 V 变化至 VDD。	-	-	50	ms
POR 完成 时间* 5	PORT		PSUP 之后至 掉电模式* 6	-	-	100	µs
电源关闭 电压* 5	SDV		关闭电压以启用 POR以重启* 6	-	-	0.2	V
电源开启 间隔* 5	PSINT		电压低于SDV所需保持的时间段，以便使POR能够保持启用以便 POR 重新启动	100	-	-	µs
模式设置前的 等待时间	Twait	-	-	100	-	-	µs

注：* 5. 设计参考值。* 6. 当上电复位电路检测到VDD电压上升时，它会复位内部电路并初始化寄存器。复位后，AK09973D将进入掉电模式。

[VDD电压波形]



8.3. AC Characteristics of OD-INT

Parameter	Symbol	Pin	Condition	Min.	Typ.	Max.	Unit
Fall time of OD-INT	TfOD	OD-INT	CL = 50 pF RL = 20 kΩ(typ.)	-	-	250	ns

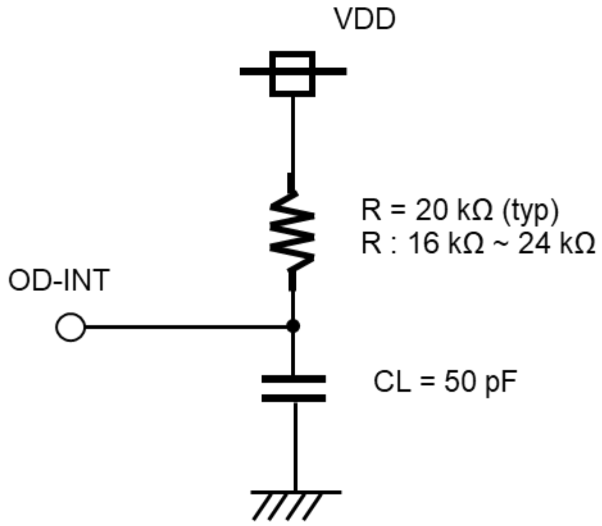
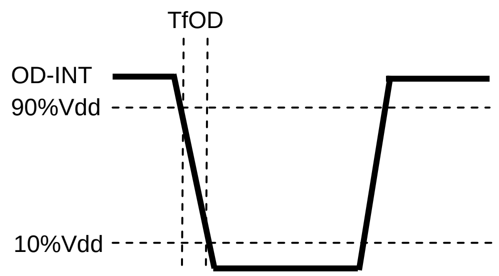


Figure 8.1 Condition of operation verification

[Rise time and fall time]



8.3. OD-INT 交流特性

参数	符号	Pin	条件	Min.	Typ.	Max.	Unit
OD-INT下降时间	TfOD	OD-INT	CL = 50 pF RL = 20 kΩ(典型值)	-	-	250	ns

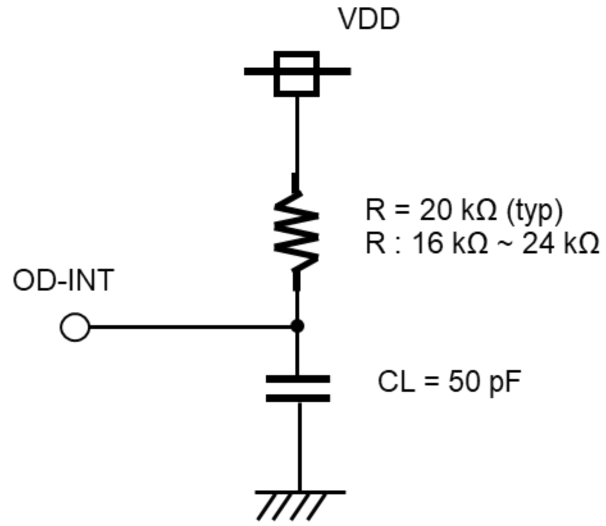
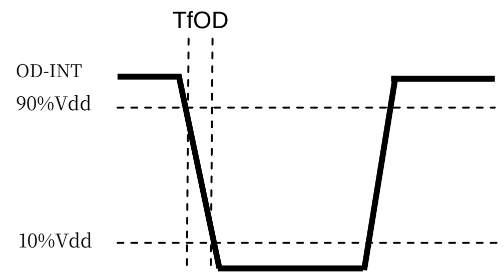


图8.1 操作验证条件

[上升时间和下降时间]



8.4. Overall Characteristics

Table 8.1 High sensitivity setting

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Measurement data output bit	DBIT	-	-	16	-	Bit
Time for measurement	TSM	SDR bit = “0” (Low noise drive)	-	0.825	0.908	ms
		SDR bit = “1” (Low power drive)	-	0.265	0.292	
Magnetic sensor sensitivity	BSE	Ta = 25˚C, SMR bit = “0”	0.99	1.1	1.21	µT/LSB
Magnetic sensor measurement range* 7	BRG	Ta = 25˚C, SMR bit = “0”	±32.44	±36.04	±39.64	mT
Magnetic sensor initial offset* 8	BOF	Ta = 25˚C	-500	-	+500	LSB
Noise* 7	NIS	SDR bit = “0” (Low noise drive)	-	5.5	-	µTrms
		SDR bit = “1” (Low power drive)	-	15.0	-	

Table 8.2 Wide range setting

Parameter	Symbol	Condition	Min.	Typ.	Max.	Unit
Measurement data output bit	DBIT	-	-	16	-	Bit
Time for measurement	TSM	SDR bit = “0” (Low noise drive)	-	0.825	0.908	ms
		SDR bit = “1” (Low power drive)	-	0.265	0.292	
Magnetic sensor sensitivity	BSE	Ta = 25˚C, SMR bit = “1”	2.79	3.1	3.41	µT/LSB
Magnetic sensor measurement range* 7	BRG	Ta = 25˚C, X and Y-axis, SMR bit = “1”	±31.42	±34.91	±38.4	mT
		Ta = 25˚C, Z-axis, SMR bit = “1”	±91.42	±101.57	±111.73	
Magnetic sensor initial offset* 8	BOF	Ta = 25˚C	-177	-	+177	LSB
Noise* 7	NIS	SDR bit = “0” (Low noise drive)	-	6.8	-	µTrms
		SDR bit = “1” (Low power drive)	-	18.0	-	

Notes:
* 7. Reference value for design. Under steady magnetic field.
* 8. Value of measurement data register on shipment test without applying magnetic field on purpose .

8.4. 整体特性

表 8.1 高灵敏度设置

参数	符号	条件	Min.	Typ.	Max.	Unit
测量数据 输出位	DBIT	-	-	16	-	Bit
测量时间	TSM	SDR位 = “0” (低噪声驱动)	-	0.825	0.908	ms
		SDR位 = “1” (低功耗驱动)	-	0.265	0.292	
磁传感器 灵敏度	BSE	环境温度 = 25℃, SMR 位 = “0”	0.99	1.1	1.21	微特斯拉/最低有效位
磁传感器 传感器测量范围* 7	BRG	环境温度 = 25℃, SMR 位 = “0”	±32.44	±36.04	±39.64	mT
磁传感器 初始偏移* 8	BOF	环境温度 = 25˚ ℃	-500	-	+500	LSB
噪声* 7	NIS	SDR位 = “0” (低噪声驱动)	-	5.5	-	微均方根
		SDR位 = “1” (低功耗驱动)	-	15.0	-	

表8.2 宽范围设置

参数	符号	条件最小值	.	Typ.	Max.	Unit
测量数据 输出位	DBIT	-	-	16	-	Bit
测量时间	TSM	SDR位 = “0” (低噪声驱动)	-	0.825	0.908	ms
		SDR位 = “1” (低功耗驱动)	-	0.265	0.292	
磁传感器 灵敏度	BSE	环境温度 = 25℃, SMR位 = “1”	2.79	3.1	3.41	微特斯拉/最低有效位
磁传感器 测量范围* 7	BRG	环境温度 = 25˚C, X轴和Y轴, 状态位 = “1”	±31.42	±34.91	±38.4	mT
		环境温度 = 25℃, Z轴, SMR位 = “1”	±91.42	±101.57	±111.73	
磁传感器 初始偏移* 8	BOF	环境温度 = 25˚ ℃	-177	-	+177	LSB
噪声* 7	NIS	SDR位 = “0” (低噪声驱动)	-	6.8	-	微均方根
		SDR位 = “1” (低功耗驱动)	-	18.0	-	

注：* 7. 设计参考值。在稳定磁场条件下。* 8. 出货测试时，在未特意施加磁场的情况下，测量数据寄存器的值。

8.5. I²C Bus Interface

I²C bus interface is compliant with Standard mode, Fast mode and Fast mode plus. As for tR and tF, specifications for Fast mode plus are applied.

■ Fast mode plus

Symbol	Parameter	Min.	Typ.	Max.	Unit
fSCL	SCL clock frequency	-	-	1000	kHz
tHIGH	SCL clock “High” time	0.26	-	-	µs
tLOW	SCL clock “Low” time	0.5	-	-	µs
tR	SDA and SCL rise time	-	-	120	ns
tF	SDA and SCL fall time	-	-	120	ns
tHD:STA	Start Condition hold time	0.26	-	-	µs
tSU:STA	Start Condition setup time	0.26	-	-	µs
tHD:DAT	SDA hold time (vs. SCL falling edge)	0	-	-	µs
tSU:DAT	SDA setup time (vs. SCL rising edge)	50	-	-	ns
tSU:STO	Stop Condition setup time	0.26	-	-	µs
tBUF	Bus free time	0.5	-	-	µs
tSP	Noise suppression pulse width	-	-	50	ns

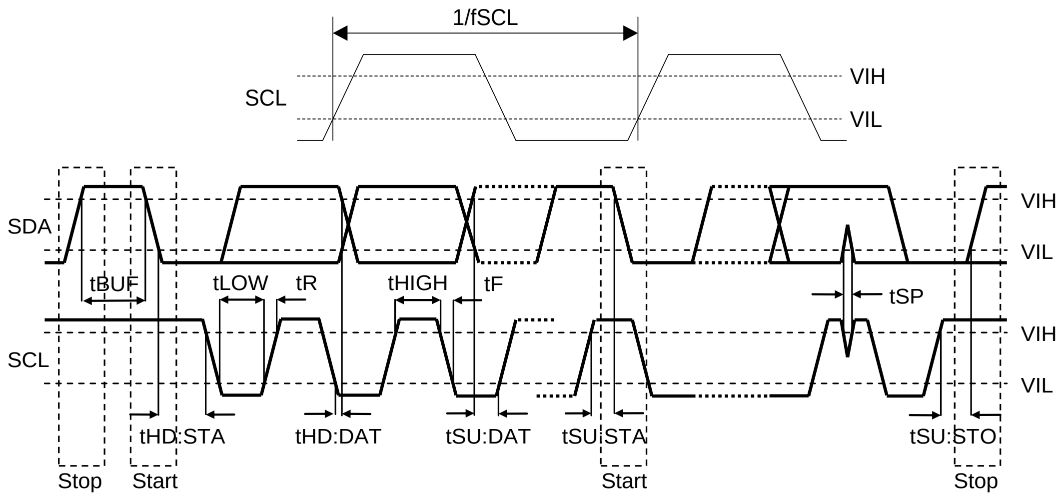


Figure 8.2 I²C bus interface timing

8.5. I²C 总线接口I²C 总线接口兼容

标准模式、快速模式和增强快速模式。关于 tR 和 tF，则采用增强快速模式的规格。

■ 增强快速模式

符号	参数	Min.	Typ.	Max.	Unit
fSCL	SCL时钟频率	-	-	1000	kHz
tHIGH	SCL时钟“高电平”时间	0.26	-	-	µs
tLOW	SCL时钟“低电平”时间	0.5	-	-	µs
tR	SDA与SCL上升时间	-	-	120	ns
tF	SDA与SCL下降时间	-	-	120	ns
tHD:STA	起始条件保持时间	0.26	-	-	µs
tSU:STA	起始条件建立时间	0.26	-	-	µs
tHD:DAT	SDA保持时间（相对于SCL下降沿）	0	-	-	µs
tSU:DAT	SDA建立时间（相对于SCL上升沿）	50	-	-	ns
tSU:STO	停止条件建立时间	0.26	-	-	µs
tBUF	总线空闲时间	0.5	-	-	µs
tSP	噪声抑制脉冲宽度	-	-	50	ns

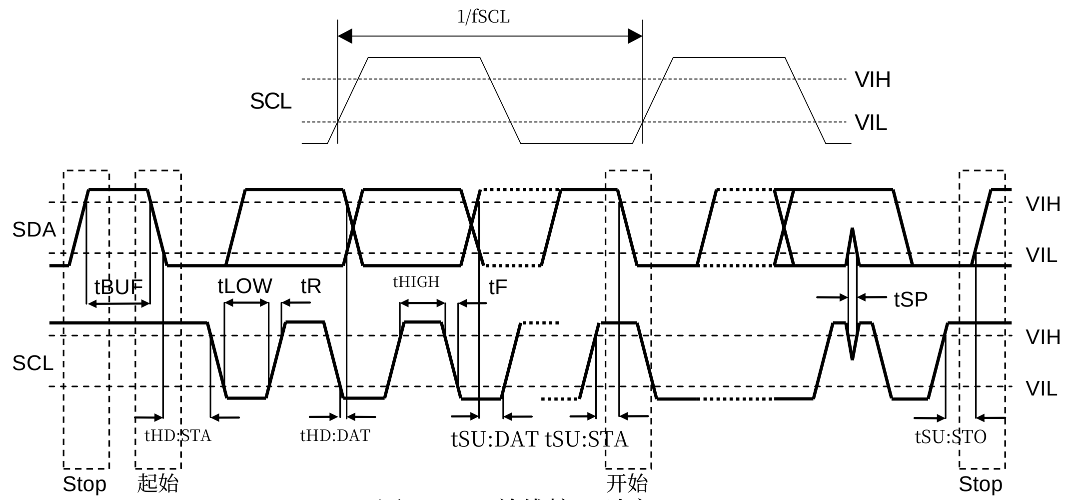
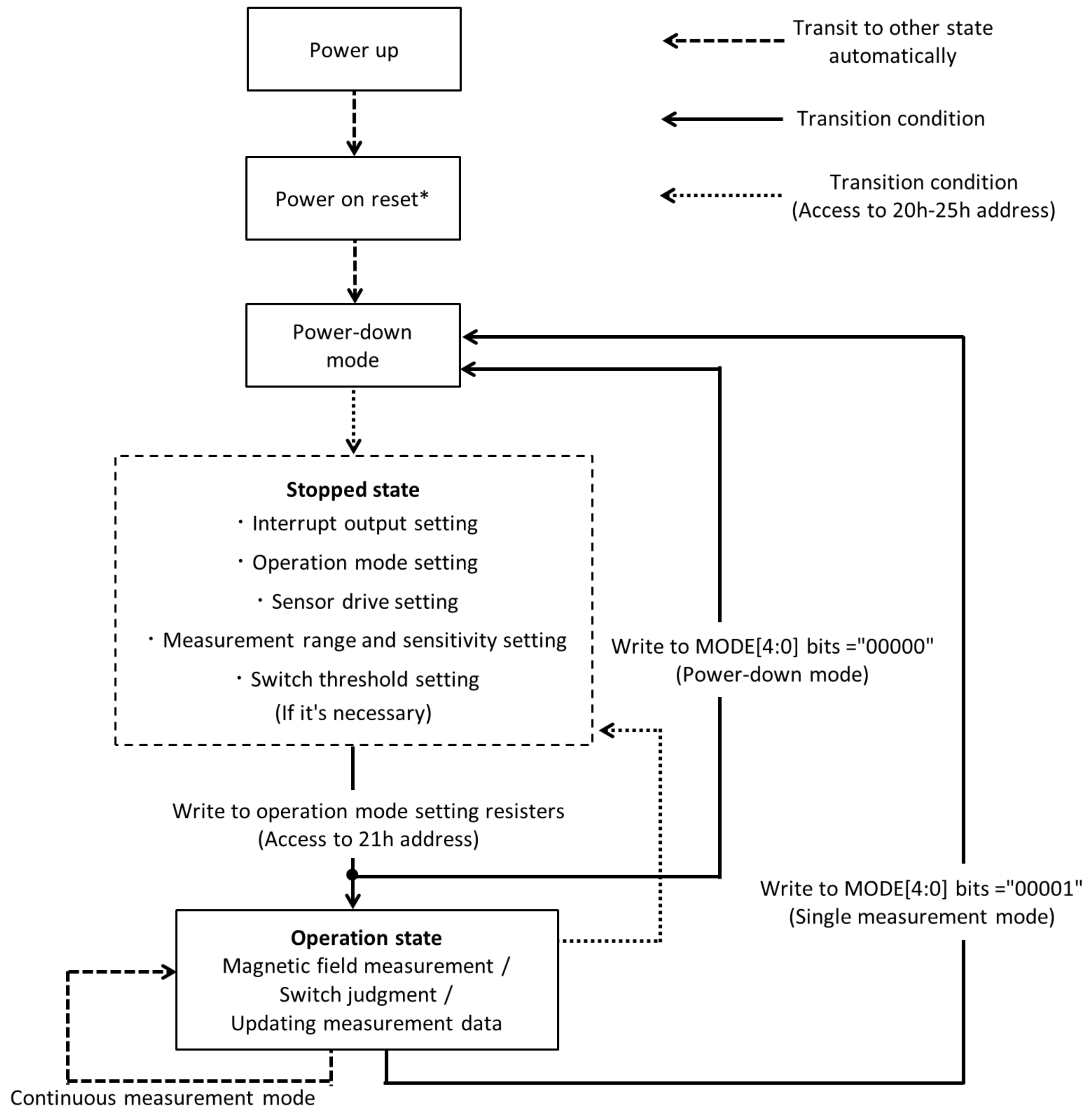


图 8.2 I²C总线接口时序

9. Status Description

9.1. State Transition Diagram

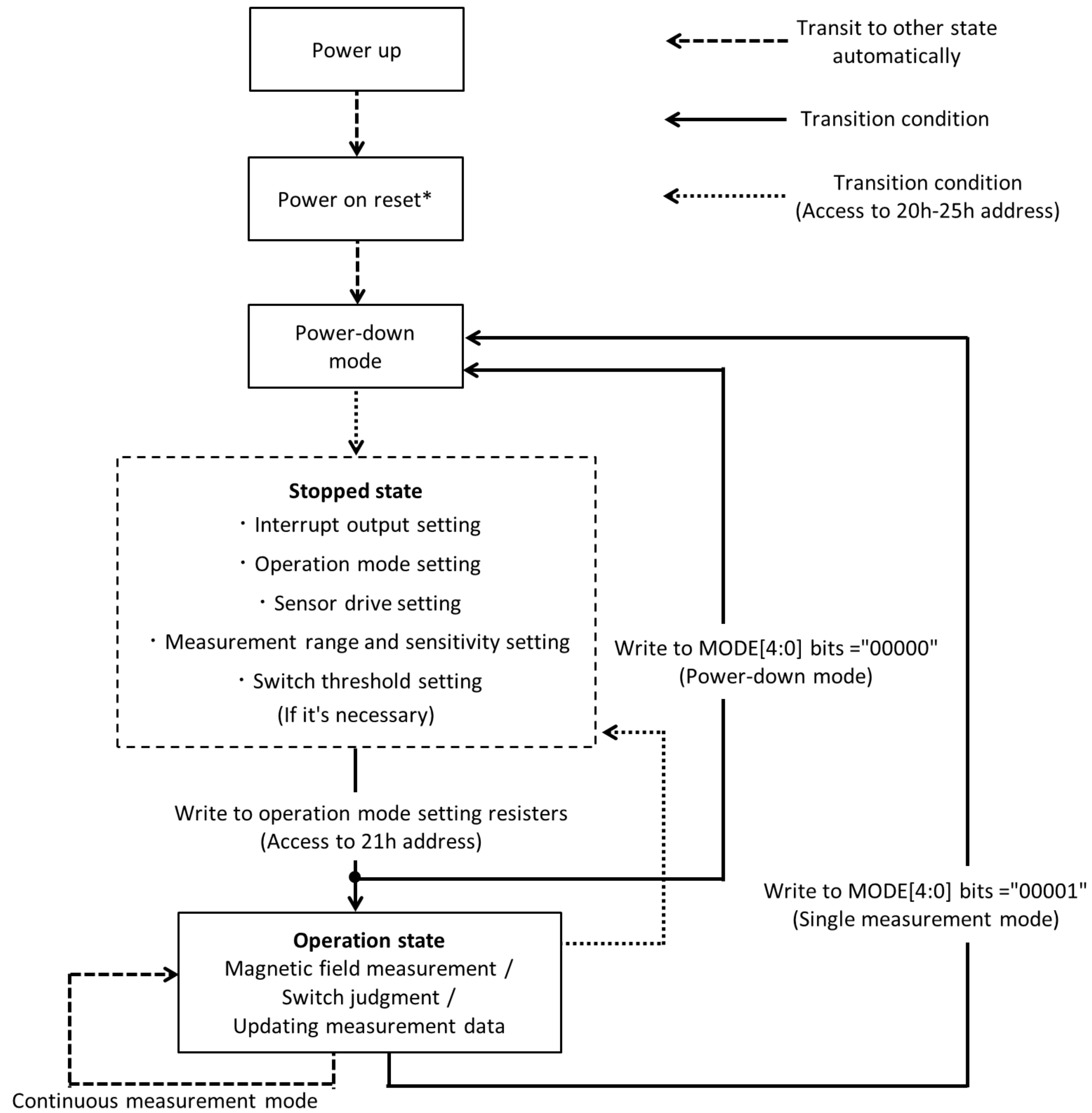


*After reset is completed, all resisters are initialized and AK09973D transits to Power-down mode automatically.

Figure 9.1 State transition diagram

9. 状态描述

9.1. 状态转换图



*After reset is completed, all resisters are initialized and AK09973D transits to Power-down mode automatically.

图 9.1 状态转换图

9.2. Power States

When VDD is turned on from Vdd = OFF (0 V), all registers in AK09973D are initialized by POR circuit and transit to Power-down mode automatically.

Table 9.1 Power States

State	VDD	Power state
1	OFF (0 V)	OFF (0 V). It does not affect external interface.
2	1.65 V to 1.95 V	ON

9.2. 电源状态当 VDD 被开启

当VDD从= 关闭（0 V）状态上电时，AK09973D中的所有寄存器均由上电复位电路初始化，并自动进入掉电模式。

表 9.1 电源状态

状态	VDD	电源状态
1	关闭 (0 V)	关闭 (0 V)。 它不影响外部接口。
2	1.65 V 至 1.95 V	开启

10. Functional Descriptions

10.1. Reset Functions

AK09973D has two types of reset;

I. Power on Reset (POR)
When Vdd rise is detected, POR circuit operates, and AK09973D is reset. After reset is completed, all registers are initialized and AK09973D transits to Power-down mode.

II. Soft reset
AK09973D is reset by setting SRST bit = “1”. After reset is completed, all registers are initialized and AK09973D transits to Power-down mode automatically.

10.2. Operation modes

AK09973D has following eleven operation modes:

(1) Power-down mode (MODE[4:0] bits = “00h,03h,05h,07h,09h,0Bh,0Dh,0Fh, ≥ 11h”)

(2) Single measurement mode (MODE[4:0] bits = “01h”)
- Sensor is measured for one time and data is output. Transits to Power-down mode (MODE[4:0] bits = “00h”) automatically after measurement ended.

(3) Continuous measurement mode 1 (MODE[4:0] bits = “02h”)
- Sensor is measured periodically in 5 Hz. Transits to other operation mode by writing MODE[4:0] bits directly.

(4) Continuous measurement mode 2 (MODE[4:0] bits = “04h”)
- Sensor is measured periodically in 10 Hz. Transits to other operation mode by writing MODE[4:0] bits directly.

(5) Continuous measurement mode 3 (MODE[4:0] bits = “06h”)
- Sensor is measured periodically in 20 Hz. Transits to other operation mode by writing MODE[4:0] bits directly.

(6) Continuous measurement mode 4 (MODE[4:0] bits = “08h”)
- Sensor is measured periodically in 50 Hz. Transits to other operation mode by writing MODE[4:0] bits directly.

(7) Continuous measurement mode 5 (MODE[4:0] bits = “0Ah”)
- Sensor is measured periodically in 100 Hz. Transits to other operation mode by writing MODE[4:0] bits directly.

(8) Continuous measurement mode 6 (MODE[4:0] bits = “0Ch”)
- Sensor is measured periodically in 500 Hz. Transits to other operation mode by writing MODE[4:0] bits directly.

(9) Continuous measurement mode 7 (MODE[4:0] bits = “0Eh”)
- Sensor is measured periodically in 1000 Hz. Transits to other operation mode by writing MODE[4:0] bits directly. This mode only enables when AK09973D is set Low power mode (SDR bit = “1”). When set this mode on Low noise mode (SDR bit = “0”), sensor is measured periodically in 500 Hz.

(10) Continuous measurement mode 8 (MODE[4:0] bits = “10h”)
- Sensor is measured periodically in 2000 Hz. Transits to other operation mode by writing MODE[4:0] bits directly. This mode only enables when AK09973D is set Low power mode (SDR bit = “1”). When set this mode on Low noise mode (SDR bit = “0”), sensor is measured periodically in 500 Hz.

(11) Self-test mode (STEST bit = “1”)
- Self-test mode is used to check if the magnetic sensor is working normally. This mode only enables when AK09973D is set Single measurement mode.

10. 功能描述

10.1. 复位功能AK09973D 有两种类型

复位类型：I. 上电复位 当检测到VDD上升时，上电复位电路工作，AK09973D被复位。复位完成后，所有寄存器被初始化，AK09973D进入掉电模式。II. 软复位 通过将SRST位设置为= “1”来复位AK09973D。复位完成后，所有寄存器被初始化，AK09973D自动进入掉电模式。

10.2. 操作模式AK09973D 具有以下操
作模式：

(1) 掉电模式 (MODE[4:0] 位 = “00h,03h,05h,07h,09h,0Bh,0Dh,0Fh, ≥ 11h”) (2) 单次测量模式 (MODE[4:0] 位 = “01h”) ➢ 传感器进行一次测量并输出数据。测量结束后自动转换到掉电模式 (MODE[4:0] 位 = “00h”)。 (3) 连续测量模式1 (MODE[4:0] 位 = “02h”) ➢ 传感器以5赫兹的频率进行周期性测量。通过直接写入MODE[4:0] 位可转换到其他操作模式。 (4) 连续测量模式2 (MODE[4:0] 位 = “04h”) ➢ 传感器以10赫兹的频率进行周期性测量。通过直接写入MODE[4:0] 位可转换到其他操作模式。 (5) 连续测量模式3 (MODE[4:0] 位 = “06h”) ➢ 传感器以20赫兹的频率进行周期性测量。通过直接写入MODE[4:0] 位可转换到其他操作模式。 (6) 连续测量模式4 (MODE[4:0] 位 = “08h”) ➢ 传感器以50赫兹的频率进行周期性测量。通过直接写入MODE[4:0] 位可转换到其他操作模式。 (7) 连续测量模式5 (MODE[4:0] 位 = “0Ah”) ➢ 传感器以100赫兹的频率进行周期性测量。通过直接写入MODE[4:0] 位可转换到其他操作模式。 (8) 连续测量模式6 (MODE[4:0] 位 = “0Ch”) ➢ 传感器以500赫兹的频率进行周期性测量。通过直接写入MODE[4:0] 位可转换到其他操作模式。 (9) 连续测量模式7 (MODE[4:0] 位 = “0Eh”) ➢ 传感器以1000赫兹的频率进行周期性测量。通过直接写入MODE[4:0] 位可转换到其他操作模式。此模式仅在AK09973D设置为低功耗模式 (SDR位 = “1”) 时启用。若在低噪声模式 (SDR位 = “0”) 下设置此模式，传感器将以500赫兹的频率进行周期性测量。 (10) 连续测量模式8 (MODE[4:0] 位 = “10h”) ➢ 传感器以2000赫兹的频率进行周期性测量。通过直接写入MODE[4:0] 位可转换到其他操作模式。此模式仅在AK09973D设置为低功耗模式 (SDR位 = “1”) 时启用。若在低噪声模式 (SDR位 = “0”) 下设置此模式，传感器将以500赫兹的频率进行周期性测量。 (11) 自检模式 (STEST位 = “1”) ➢ 自检模式用于检查磁传感器是否正常工作。此模式仅在AK09973D设置为单次测量模式时启用。

10.2.1. Description of Each Operation Mode

10.2.1.1. Power-down Mode

Power to almost all internal circuits is turned off, all registers are accessible in Power-down mode and data stored in read/write registers are remained. They can be reset by reset function.

10.2.1.2. Single Measurement Mode

When Single measurement mode (MODE[4:0] bits = “01h”) is set, magnetic sensor measurement is started. After magnetic sensor measurement and signal processing is finished, measurement magnetic data is stored to measurement data registers (HX, HY, HZ and HV registers), then AK09973D transits to Power-down mode automatically. On transition to Power-down mode, MODE[4:0] bits turns to “0”. At the same time, DRDY bit in ST register turns to “1” and SW bits in ST register turn to another state when measurement magnetic data exceed a setup threshold value.

10.2.1.3. Continuous Measurement Mode 1,2,3,4,5,6,7 and 8

When Continuous measurement modes (1 to 8) are set, magnetic sensor measurement is started periodically at 5 Hz, 10 Hz, 20 Hz, 50 Hz, 100 Hz, 500 Hz, 1000 Hz and 2000 Hz respectively. After magnetic sensor measurement and signal processing is finished, measurement magnetic data is stored to measurement data registers and all circuits except for the minimum circuit required for counting cycle length are turned off (Power Save: PS). When the next measurement timing comes, AK09973D wakes up automatically from PS and starts measurement again.

Continuous measurement mode ends when a different operation mode is set. When user access to Setting Registers (address 20h to 25h), AK09973D stops updating switch states and measurement data registers.

Table 10.1 Continuous measurement modes

Operation mode	Register setting (MODE[4:0] bits)	Measurement frequency [Hz]
Continuous measurement mode 1	0 0010	5
Continuous measurement mode 2	0 0100	10
Continuous measurement mode 3	0 0110	20
Continuous measurement mode 4	0 1000	50
Continuous measurement mode 5	0 1010	100
Continuous measurement mode 6	0 1100	500
Continuous measurement mode 7	0 1110	1000
Continuous measurement mode 8	1 0000	2000

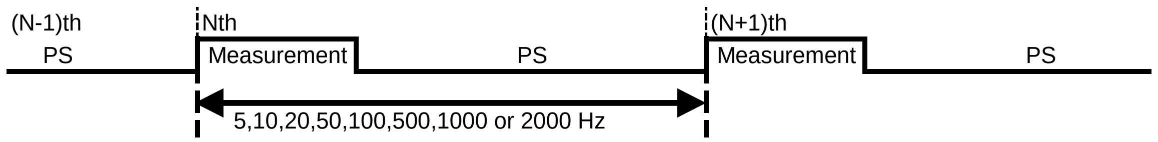


Figure 10.1 Continuous measurement modes

10.3. Data Ready

When measurement data is stored and ready to be read, DRDY bit in ST register turns to “1”. This is called “Data Ready”. When DRDYEN bit in CNTL1 register is “1”, OD-INT pin notify user of the Data Ready state. When any of measurement data register (HX,HY,HZ and HV register) is read all the way through or access to Setting Registers (address 20h to 25h), DRDY bit turns to “0”.

10.2.1. 各操作模式描述 10.2.1.1. 掉电模式 几

乎所有内部电路的电源均被关闭

，在掉电模式下所有寄存器均可访问，且读写寄存器中存储的数据保持不变。它们可通过复位功能进行复位。

10.2.1.2. 单次测量模式 当单次测量模式（MO

DE[4:0] 位 = “01h”）被设置时，磁传感器测量启动。磁传感器测量和信号处理完成后，测量磁数据被存储到测量数据寄存器（HX、HY、HZ和HV寄存器）中，随后AK09973D自动转换到掉电模式。在转换到掉电模式时，MODE[4:0] 位变为“0”。同时，状态寄存器中的DRDY位变为“1”，并且当测量磁数据超过设定的阈值时，状态寄存器中的SW位会转变为另一种状态。

10.2.1.3. 连续测量模式1、2、3、4、5、6、7和8 当设置连续测量模式（1至8）时，磁

传感器测量将分别以5赫兹、10赫兹、20赫兹、50赫兹、100赫兹、500赫兹、1000赫兹和2000赫兹的频率周期性启动。磁传感器测量和信号处理完成后，测量磁数据将被存储到测量数据寄存器中，并且除维持周期计数所需的最小电路外，所有电路都将关闭（省电模式：PS）。当下一个测量时刻到来时，AK09973D会自动从PS唤醒并再次开始测量。

当设置了不同的操作模式时，连续测量模式结束。当用户访问设置寄存器（地址20h至25h）时，AK09973D将停止更新开关状态和测量数据寄存器。

表 10.1 连续测量模式

操作模式	寄存器设置 (模式[4:0] 位)	测量 频率 [赫兹]
连续测量模式1	0 0010	5
连续测量模式2	0 0100	10
连续测量模式3	0 0110	20
连续测量模式4	0 1000	50
连续测量模式5	0 1010	100
连续测量模式6	0 1100	500
连续测量模式7	0 1110	1000
连续测量模式8	1 0000	2000

图 10.1 连续测量模式

10.3. 数据就绪当

测量数据已存储并准备就绪可读取时，状态寄存器中的DRDY位会变为“1”。这被称为“数据就绪”。当CNTL1寄存器中的DRDYEN位为“1”时，OD-INT引脚会向用户通知数据就绪状态。当任一测量数据寄存器（HX、HY、HZ和HV寄存器）被完全读取或访问设置寄存器（地址20h至25h）后，DRDY位会变为“0”。

10.3.1. Normal Measurement Data Read Sequence

(1) Check Data Ready or not by any of the following method.
Monitor OD-INT pin
Polling DRDY bit of ST register

When Data Ready, proceed to the next step.

(2) Read ST and measurement data
When ST register and any of measurement data register (HX, HY, HZ and HV register) is read all the way through, or access to Setting Registers (address 20h to 25h), AK09973D judges that data reading is finished. When data reading is finished, DRDY bit and DOR bit turns to “0”.

When measurement data register is accessed, AK09973D judges that data reading is started. Stored measurement data is protected during data reading and data is not updated. By reading measurement data register is finished, this protection is released.

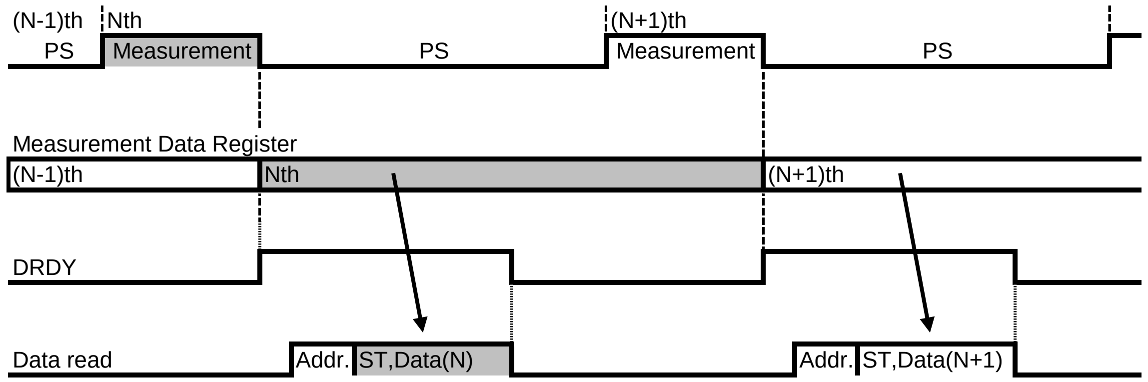


Figure 10.2 Timing chart of Measurement data read

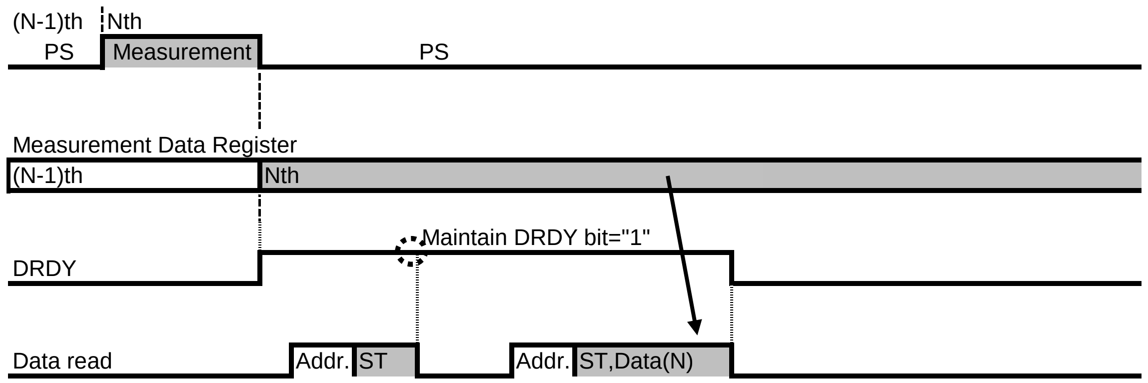


Figure 10.3 Timing chart of ST data read

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10.3.1. 正常测量数据读取序列 (1) 通过以下任一方式检查数据就绪状态

方法。监视OD-INT引脚 轮询状态寄存器的DRDY位

当数据就绪时，继续下一步。

(2) 读取状态和测量数据
当状态寄存器以及任一测量数据寄存器（HX寄存器、HY寄存器、HZ寄存器和HV寄存器）被完整读取，或访问了设置寄存器（地址20h至25h）时，AK09973D会判定数据读取已完成。数据读取完成后，DRDY位和DOR位将变为“0”。

当访问测量数据寄存器时，AK09973D会判定数据读取已开始。在数据读取期间，存储的测量数据受到保护，数据不会被更新。读取测量数据寄存器完成后，此保护即被解除。

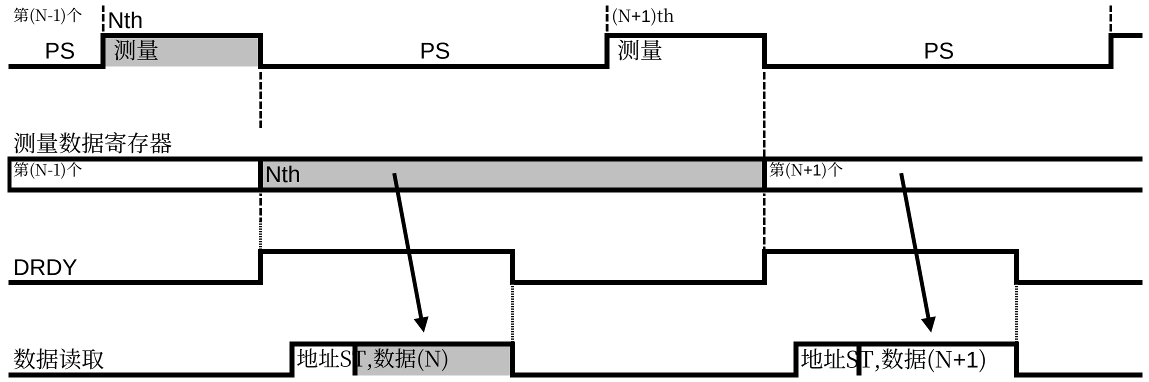

图 10.2 测量数据读取时序图

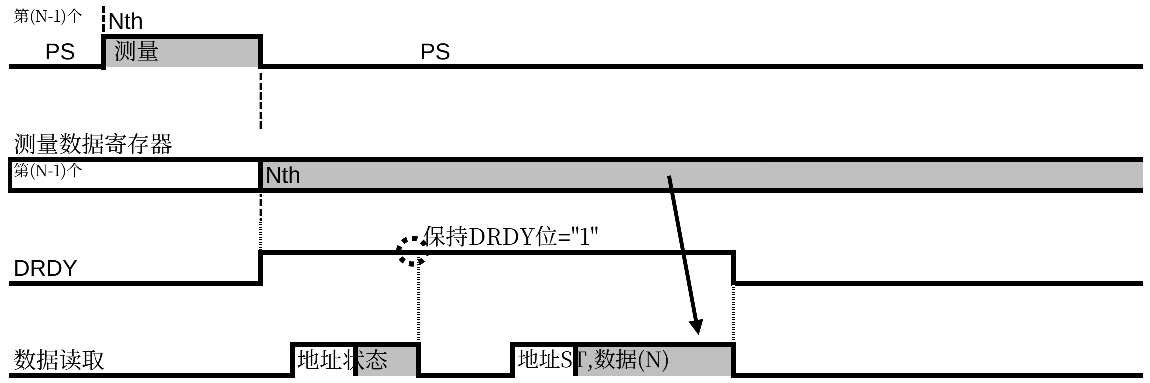

图 10.3 状态数据读取时序图

10.3.2. Data Read Start during Measurement

When the sensor is measuring (Measurement period), measurement data registers (HX, HY, HZ and HV register) keep the previous data. Therefore, it is possible to read out data even during measurement period. If data is started to be read during measurement period, previous data is read.

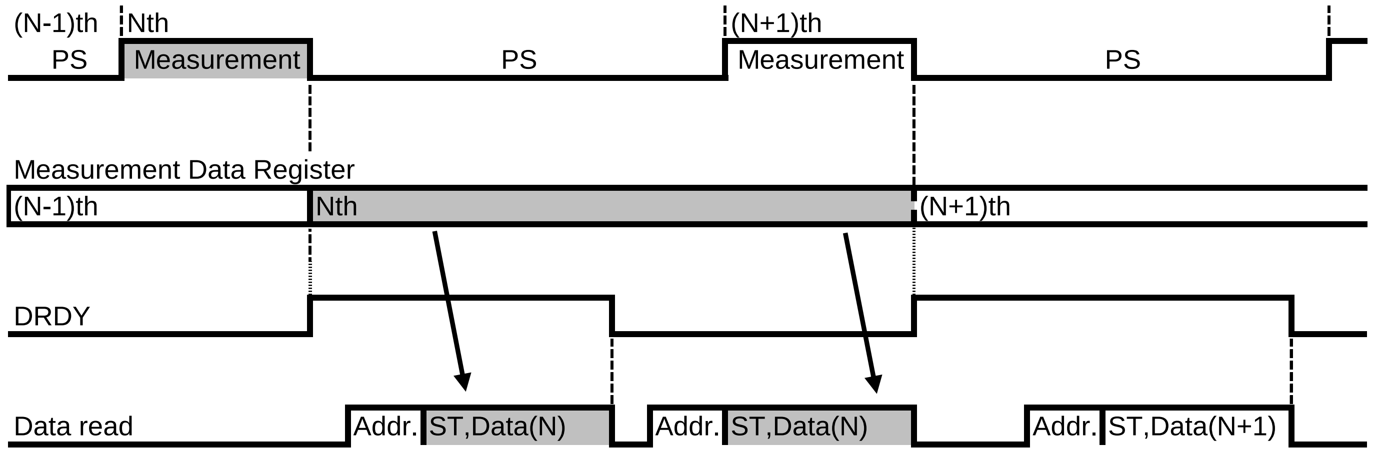

Figure 10.4 Data read start during measuring

10.3.3. Data Skip

When Nth data was not read before (N+1)th measurement ends, Data Ready remains until data is read. In this case, a set of measurement data is skipped so that DOR bit turns to "1". DOR bit turns to "0" at the (N+2)th measurement ended.

When data reading started after Nth measurement ended and did not finish reading before (N+1)th measurement ended, Nth measurement data is protected to keep correct data. In this case, a set of measurement data is not skipped and stored after finish reading Nth measurement data so that DOR bit = "0".

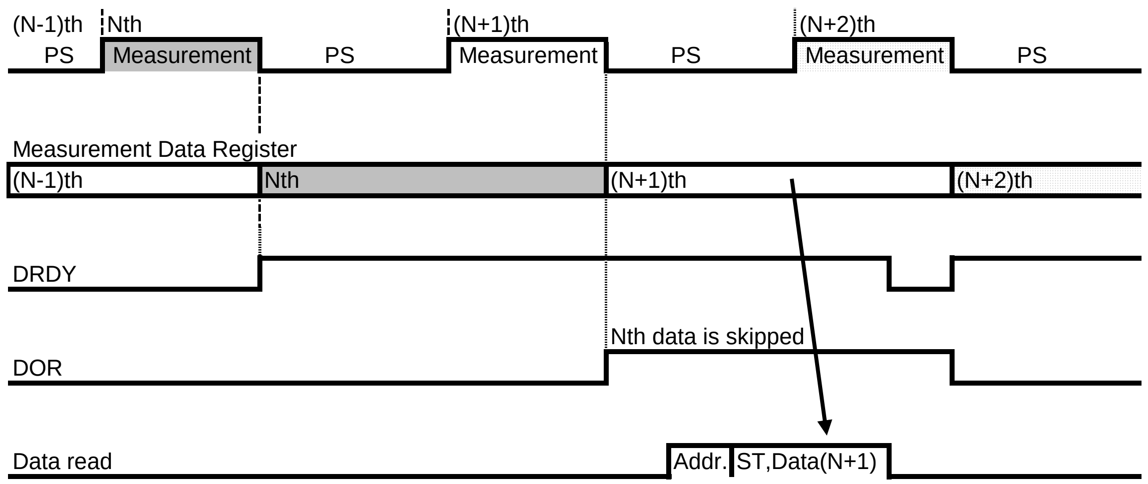

Figure 10.5 Data Skip: When data is not read

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10.3.2. 测量期间数据读取开始当传感器正在测量时（测量期间

测量周期内，测量数据寄存器（HX、HY、HZ和HV寄存器）会保持先前数据。因此，即使在测量周期内也可以读取数据。如果在测量周期内开始读取数据，读取到的将是先前数据。

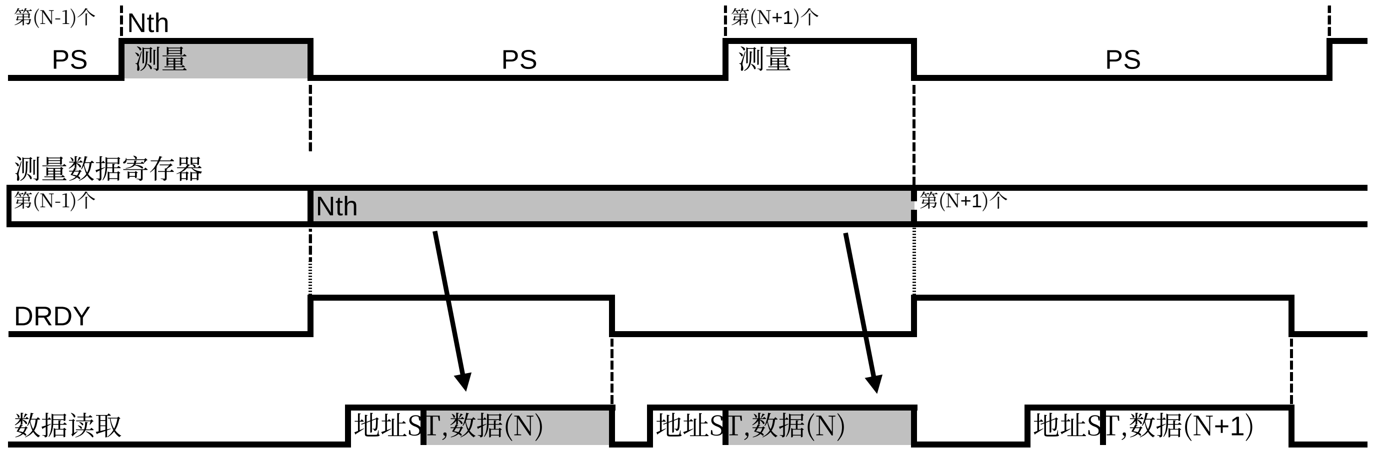

图 10.4 测量期间的数据读取开始

10.3.3. 当第N次数据未读取时的数据跳过

如果在第+1次测量结束前未读取数据，数据就绪状态将保持，直到数据被读取。在这种情况下，会跳过一组测量数据，导致数据溢出位变为"1"。 数据溢出位在第次测量结束时变为"0"。
当数据读取在第N次测量结束后开始，且在第(N+1)次测量结束前未完成读取时，第N次测量数据会受到保护以保持数据正确。在这种情况下，一组测量数据不会被跳过，并在完成读取第N次测量数据后存储，从而使数据溢出位= "0"。

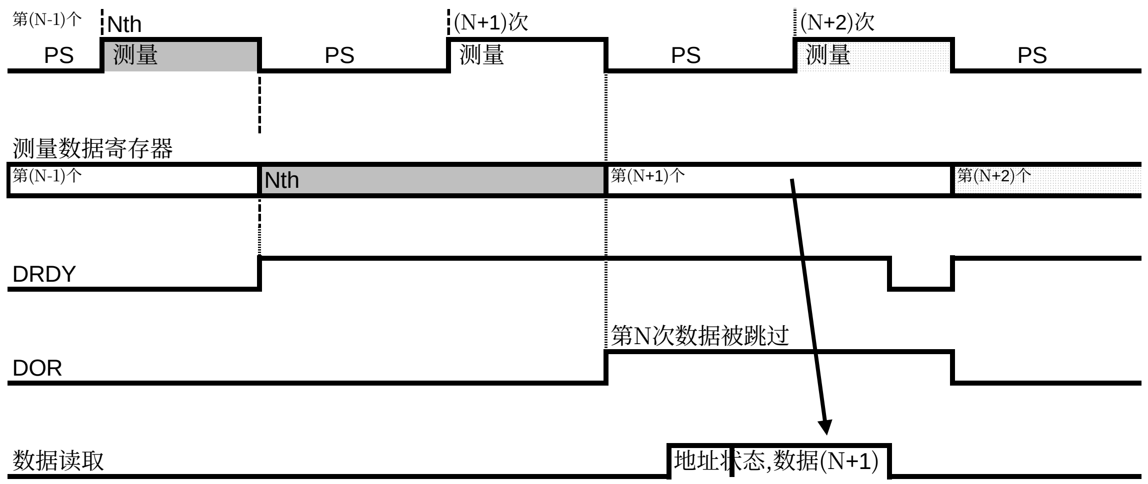

图 10.5 数据跳过：当数据未被读取时

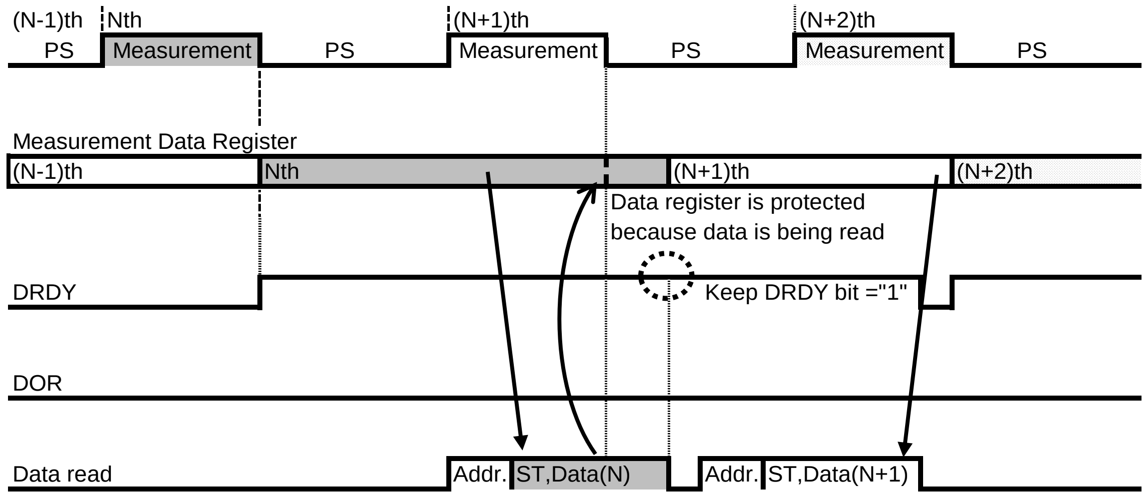

Figure 10.6 Data Not Skip: When data read has not been finished before the next measurement end

10.3.4. End Operation

Set Power-down mode (MODE[4:0] bits = “00h”) to end Continuous measurement mode.

10.4. Programmable Switch Function

AK09973D has a programmable switch function created by setting switch threshold values (operating threshold* [9] and returning threshold* [10]) and switch function enable bits (SWEN bits* [11]). When measurement magnetic data exceeds the operating threshold value, switch event bit (SW bits* [12]) turns to “1”. When measurement magnetic data is lower than the returning threshold, SW bits turns to “0”. The switch function is used to check the magnitude relation between the measurement data and the switch threshold values. After the magnetic sensor measurement and signal processing has finished, measurement data is stored to the measurement data register. Then AK09973D compares the measurement data with the defined switch threshold values and outputs the comparison results at the SW bits in ST register. Switch thresholds can be free to set (Settable range: same as measurement range. Settable sensitivity: same as measurement sensitivity).

Notes:
* 9. BOPX[15:0], BOPY[15:0], BOPZ[15:0] and BOPV[15:0]
* 10. BRPX[15:0], BRPY[15:0], BRPZ[15:0] and BRPV[15:0]
* 11. SWXEN bit, SWYEN bit, SWZEN bit and SWVEN bit
* 12. SWX bit, SWY bit, SWZ bit and SWV bit

Table 10.2 Relation between threshold values and SW bit of X-axis* [13]

Relation between BOPX and BRPX	Magnitude relation between measurement data and threshold values	SWX bit result
BOPX ≤ BRPX (Switch function disable)	Don’t care	Don’t care
BOPX > BRPX (Switch function enable)	BOPX < HX	1
	BRPX > HX	0
	BRPX ≤ HX ≤ BOPX	Previous result

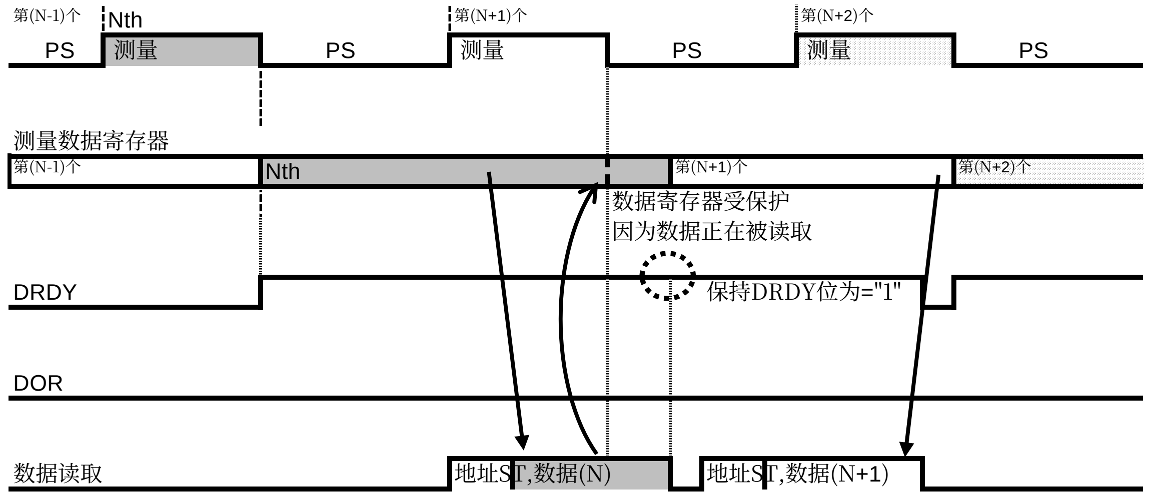

图10.6 数据未跳过：当数据读取在下一次测量结束前尚未完成时

10.3.4. 结束操作设置掉电模式（

模式[4：0] 位 = “00h”）以结束连续测量模式。

10.4. 可编程开关功能AK09973D 具有一个可编程开关功

能，通过设置开关阈值（工作阈值* [9] 和返回阈值* [10]）以及开关功能使能位（SWEN位* [11]）来创建。当测量磁数据超过工作阈值时，开关事件位（SW位* [12]）变为“1”。当测量磁数据低于返回阈值时，SW位变为“0”。该开关功能用于检查测量数据与开关阈值之间的大小关系。磁传感器测量和信号处理完成后，测量数据被存储到测量数据寄存器中。然后，AK09973D 将测量数据与定义的开关阈值进行比较，并将比较结果输出到状态寄存器中的SW位。切换阈值可以自由设置（可设置范围：与测量范围相同。可设置灵敏度：与测量灵敏度相同）。

注释：* 9. BOP

X[15:0], BOPY[15:0], BOPZ[15:0] 和 BOPV[15:0]* 10. BRPX[15:0], BRPY[15:0], BRPZ[15:0] 和 BRPV[15:0]* 11. SWXEN位、SWYEN位、SWZEN位和SWVEN位 * 12. SWX位、SWY位、SWZ位和SWV位

表10.2 X轴*阈值与SW位的关系 [13]

之间的关系 BOPX 和 BRPX	大小关系介于 大小关系 数值	SWX位 结果
BOPX ≤ BRPX (开关功能禁用)	无关”	不用关心
BOPX > BRPX (开关功能 启用)	BOPX < HX寄存器	1
	BRPX > HX寄存器	0
	BRPX ≤ HX寄存器 ≤ BOPX	先前结果

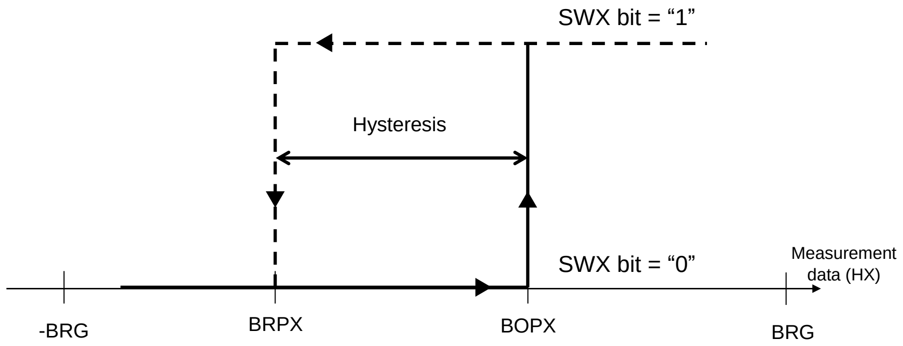

Figure 10.7 Relation between threshold values and SW bit of X-axis*13

Note:
* 13. X-axis, Y-axis, Z-axis and sum of squares of 3-axis exhibits the same relationship

10.5. Self-test Function

Self-test mode is used to check if the magnetic sensor is working normally.
When Self-test mode (STEST bit = “1”) and Single measurement mode (MODE[4:0] bits = “01h”) are set, magnetic field is generated by the internal magnetic source and magnetic sensor is measured. In the Self-test mode, the settings of Sensor drive select (SDR bit) and Sensor measurement range (SMR bit) are invalid and measurement is performed with Low noise (SDR bit = “0”) and High sensitivity (SMR bit = “0”). Measurement data is stored to measurement data registers (HX, HY, HZ), then AK09973D transits to Power-down mode automatically.
Data read sequence and functions of read-only registers in Self-test mode is the same as Single measurement mode.
When measurement data read by the self-test sequence is in the range of following table, AK09973D is working normally.

	HX[15:0] bits	**HY[15:0] bits**	**HZ[15:0] bits**
Criteria	-120 ≤ HX ≤ 120	-120 ≤ HY ≤ 120	60 ≤ HZ ≤ 400

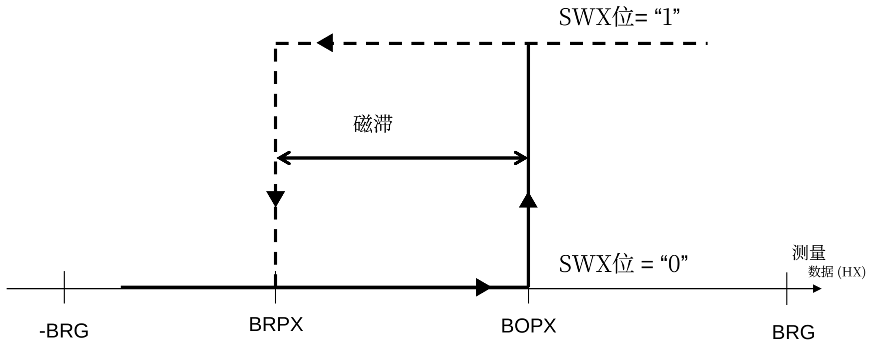

图 10.7 X轴阈值与SW位的关系*13

注意:
* 13. X轴、Y轴、Z轴以及三轴平方和均呈现相同的关系

10.5. 自检功能自检模式用于检
检查磁传感器是否正常工作。
当自检模式（STEST位 = “1”）和单次测量模式（MODE[4:0] 位 = “01h”）被设置时，内部磁源会产生磁场并对磁传感器进行测量。在自检模式下，传感器驱动选择（SDR位）和传感器测量范围（SMR位）的设置无效，测量将以低噪声（SDR位 = “0”）和高灵敏度（SMR 位 = “0”）进行。测量数据将存储到测量数据寄存器（HX、HY、HZ）中，随后AK09973D会自动转换至掉电模式。

自检模式下的数据读取序列和只读寄存器的功能与单次测量模式相同。

当通过自检序列读取的测量数据处于下表范围内时，AK09973D 工作正常。

	HX**[15:0]** 位	HY**[15:0]** 位	HZ**[15:0]** 位
标准	-120 ≤ HX寄存器 ≤ 120	-120 ≤ HY寄存器 ≤ 120	60 ≤ 赫兹 ≤ 400

10.6. Error Notification Function

AK09973D has a limitation for measurement range, where the absolute value of X-axis and Y-axis should be smaller than 36.04 mT (High sensitivity mode) or 34.91 mT (Wide range mode) and the absolute value of Z-axis should be smaller than 36.04 mT (High sensitivity mode) or 101.57 mT (Wide range mode). When the magnetic field exceeds this limitation, AK09973D outputs limitation value (fixed value: 36.04 mT, 34.91 mT or 101.57 mT) at the X-axis or/and Y-axis or/and Z-axis. This is called magnetic sensor overflow. When magnetic sensor overflow occurs, ERR bit turns to “1”. When the magnetic field less than limitation value, measurement data register and ERR bit are updated.

10.7. Interrupt Function

AK09973D has Open-drain interrupt pin (OD-INT pin). When CNTL1 register is set and interrupt event occurred, AK09973D outputs selected interrupt event at OD-INT pin. AK09973D can output three type of interrupt events (Switch event, Data ready, Error event) to OD-INT pin. Switch event occurs when measurement data is higher than BOP value and POL bit*14 = “0” or when measurement data is lower than BRP value and POL bit = “1”. When interrupt Switch event or Data ready or Error event occurs, OD-INT pin turns to “L”.

Note:
*14. POLX bit, POLY bit, POLZ bit and POLV bit

Table 10.3 Relation between threshold values of X-axis and OD-INT pin* 15

Relation between BOPX and BRPX	Magnitude relation between measurement data and threshold values	OD-INT pin	
		POLX = “0”	POLX = “1”
BOPX ≤ BRPX (Switch function disable)	Don’t care	Don’t care	
BOPX > BRPX (Switch function enable)	BOPX < HX	L	H
	BRPX > HX	H	L
	BRPX ≤ HX ≤ BOPX	Previous result	

Note:
* 15. X-axis, Y-axis, Z-axis and sum of squares of 3-axis exhibits the same relationship

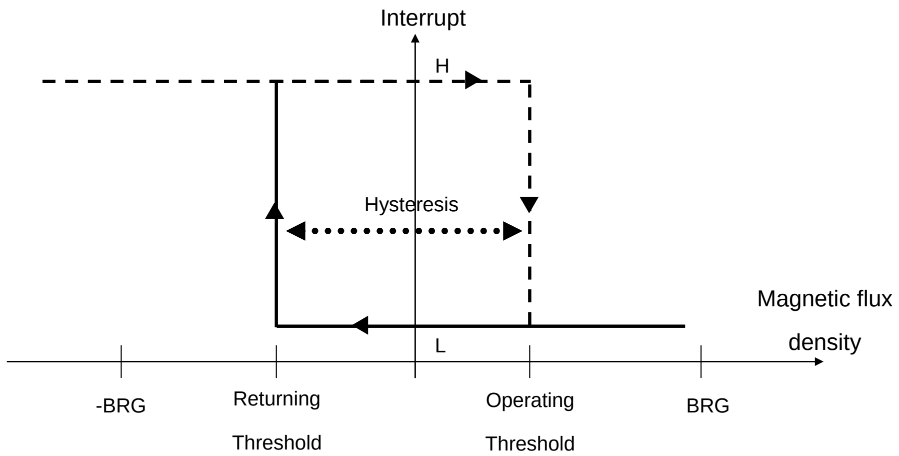

Figure 10.8 Open drain interrupt pin (POL bit = “0”)

10.6. 错误通知功能AK09973D 在测量方面存在限制

量程范围，其中X轴和Y轴的绝对值应小于36.04毫特斯拉（高灵敏度模式）或34.91毫特斯拉（宽范围模式），Z轴的绝对值应小于36.04毫特斯拉（高灵敏度模式）或101.57毫特斯拉（宽范围模式）。当磁场超过此限值时，AK09973D会在X轴和/或Y轴和/或Z轴输出限值（固定值：36.04毫特斯拉、34.91毫特斯拉或101.57毫特斯拉）。这称为磁传感器溢出。当发生磁传感器溢出时，ERR 位变为“1”。当磁场小于限值时，测量数据寄存器和ERR位会更新。

10.7. 中断功能AK09973D 具有开漏

中断引脚（OD-INT引脚）。当CNTL1寄存器被设置且中断事件发生时，AK09973D会在OD-INT引脚输出选定的中断事件。AK09973D可以向OD-INT引脚输出三种类型的中断事件（切换事件、数据就绪、错误事件）。当测量数据高于BOP值且POL位*14 = “0”，或当测量数据低于BRP值且POL位= “1”时，会发生切换事件。当中断切换事件、数据就绪或错误事件发生时，OD-INT引脚会变为“低电平”。

注:
*14. POLX位、POLY位、POLZ位和POLV位

表10.3 X轴阈值与OD-INT引脚的关系* 15

关系 BOPX与BRPX	大小关系 测量数据与阈值 值	OD-INT引脚	
		POLX = “0”	POLX = “1”
BOPX ≤ BRPX (开关功能禁用)	不’在乎	不’在乎	
BOPX > BRPX (开关功能 启用)	BOPX < HX寄存器	L	H
	BRPX > HX寄存器	H	L
	BRPX ≤ HX寄存器 ≤ BOPX	先前结果	

注意：* 15. X轴、Y轴、Z轴以及三轴平方和呈现相同的关系

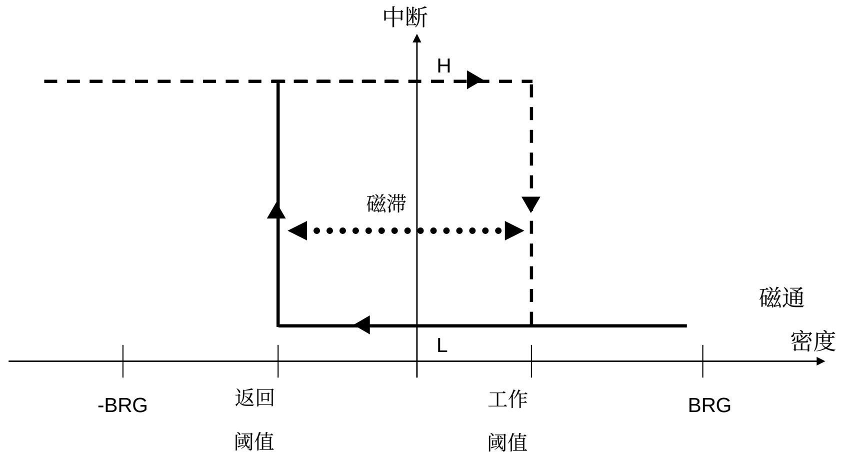

图 10.8 开漏中断引脚（POL位 = “0”）

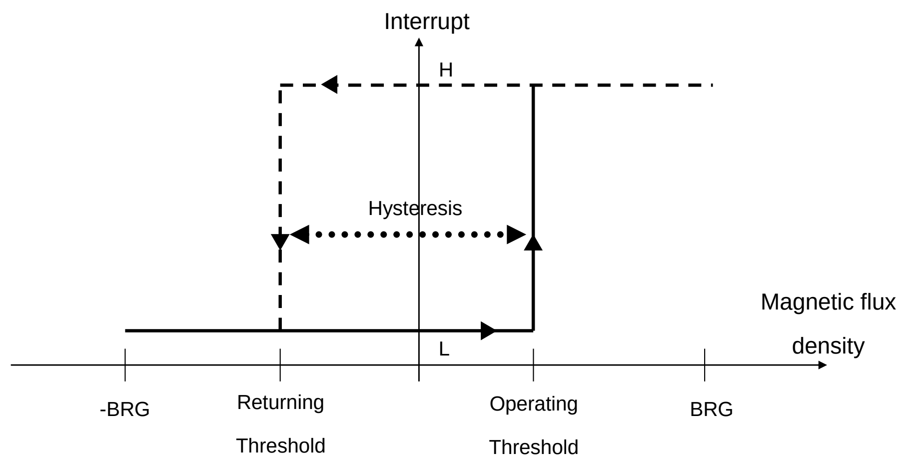

Figure 10.9 Open drain interrupt pin (POL bit = “1”)

10.7.1. Interrupt Event

(1) Switch interrupt event
- When measurement magnetic data exceeds the operating threshold value and POL bit = “0”, SW bit turns to “1” and OD-INT pin turns to “L”. When measurement magnetic data is lower than the returning threshold and POL bit = “0”, SW bit turns to “0” and OD-INT pin turns to “H”. In case of POL bit = “1”, the polarity of OD-INT pin is the reverse of when POL bit = “0”.

(2) Data ready
- OD-INT pin notifies user of the Data Ready state. When Data ready is occurred, DRDY bit turns to “1” and OD-INT pin turns to “L”. When user accesses to register address, OD-INT pin turns to “H”.

(3) Error event (Overflow)
- When magnetic sensor overflow occurs, ERR bit turns to “1” and OD-INT pin turns to “L”. When the magnetic field less than limitation value, ERR bit turns to “0” and OD-INT pin turns to “H”.

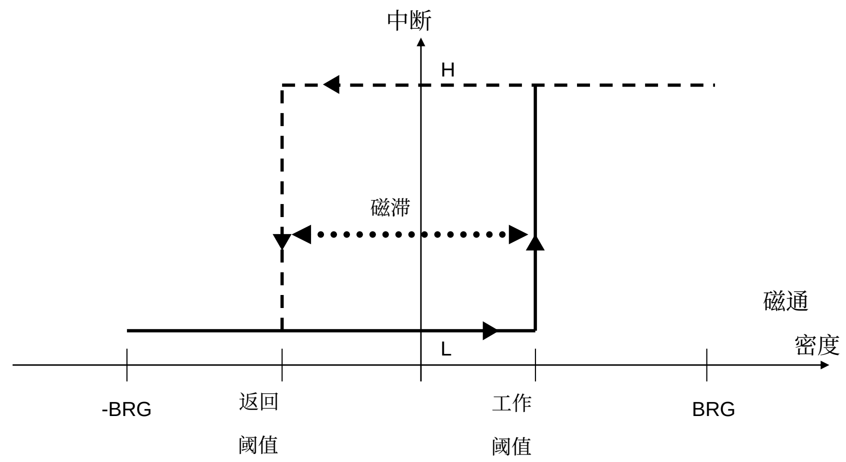

图 10.9 开漏中断引脚 (POL位 = “1”)

10.7.1. 中断事件

(1) 开关中断事件➢ 当测量磁数据超过工作阈值且POL位= 为“0”时，SW位变为“1”且OD-INT引脚变为“低电平”。当测量磁数据低于返回阈值且POL位= 为“0”时，SW位变为“0”且OD-INT引脚变为“高电平”。若POL位= 为“1”，则OD-INT引脚的极性与POL位= 为“0”时相反。 (2) 数据就绪 ➢ OD-INT引脚向用户通知数据就绪状态。当数据就绪发生时，DRDY位变为“1”且OD-INT引脚变为“低电平”。当用户访问寄存器地址时，OD-INT引脚变为“高电平”。 (3) 错误事件（溢出）➢ 当磁传感器溢出发生时，ERR位变为“1”且OD-INT引脚变为“低电平”。当磁场小于限值时，ERR位变为“0”且OD-INT引脚变为“高电平”。

10.7.2. Timing of DRDY Interrupt Function Operation

Timing of interrupt function operation is given below.

Table 10.3 Timing of interrupt function operation

Pin name	Output transition	Timing of transition	Remarks
OD-INT pin	H → L	End of measurement	-
	L → H	Read address 10h - 1Fh or Write address 20h - 25h	During access to address, OD-INT pin is always “H” state.

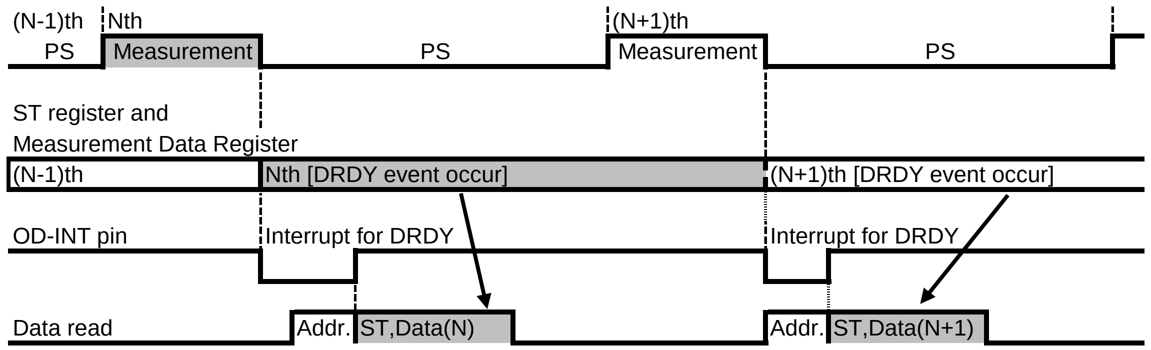

Figure 10.10 Timing chart of DRDY interrupt function (Normal read sequence)

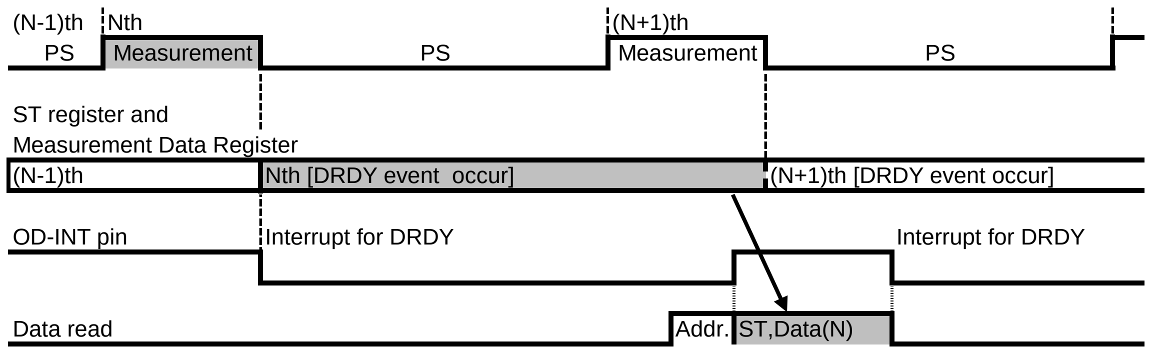

Figure 10.11 Timing chart of DRDY interrupt function
(When Nth data is read start immediately before (N+1)th measurement end)

10.7.2. DRDY中断功能操作时序 中断功能操作时序如下所示。

表10.3 中断功能操作时序

引脚名称	输出转换	转换时序	备注
OD-INT引脚	H → L	测量结束	-
	L → H	读取地址 10h - 1Fh or 写入地址 20h - 25h	在访问 地址期间，OD-INT引脚 始终处于“高电平”状态。

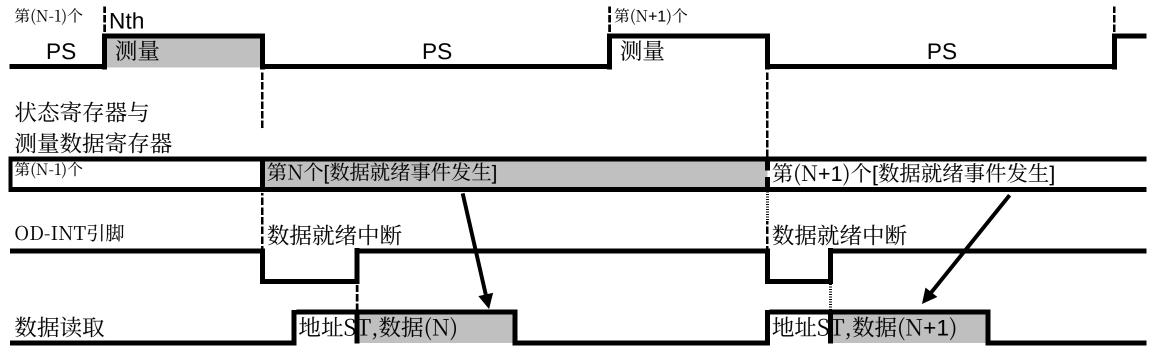

图 10.10 DRDY中断功能时序图（正常读取序列）

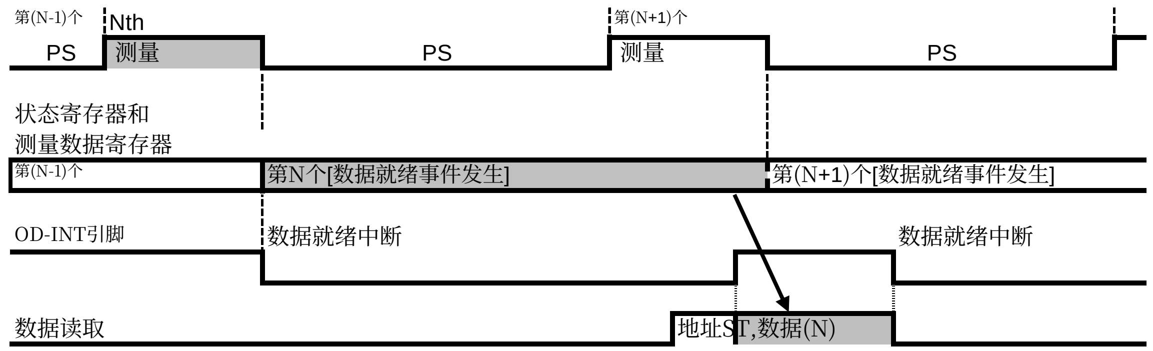

图 10.11 DRDY中断功能时序图（当第N次数据读取开始紧邻第(N+1)次测量结束时）

10.7.3. Timing of Switch/Error Interrupt Function Operation

When user assigns OD-INT pin to SW event output or/and Error event output, OD-INT pin notifies user of these event. Timing of these interrupt function operation is given below.

Table 10.5 Timing of SW/ERROR interrupt function operation (POL bit = “0”)

Pin name	Output transition	Timing of transition
OD-INT pin	H → L	End of measurement(SW/ERROR)
	L → H	End of measurement(SW/ERROR) or Write address 20h - 25h

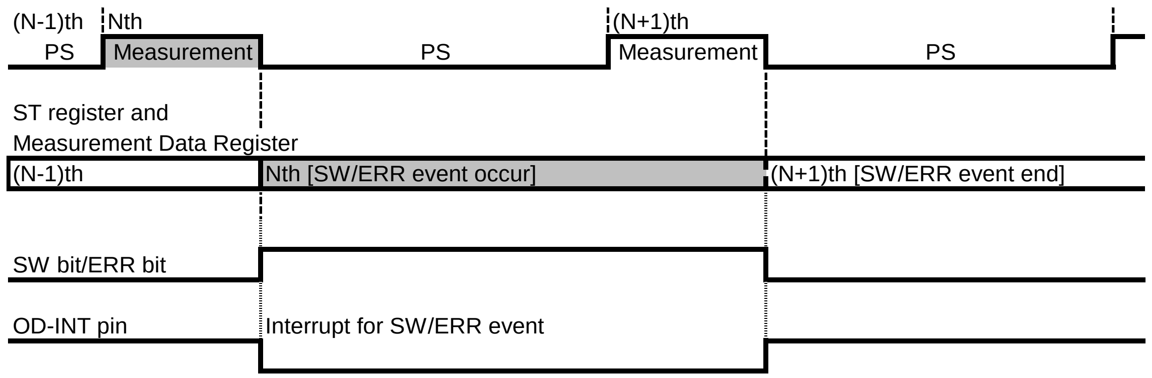

Figure 10.12 Timing chart of SW/ERROR interrupt function

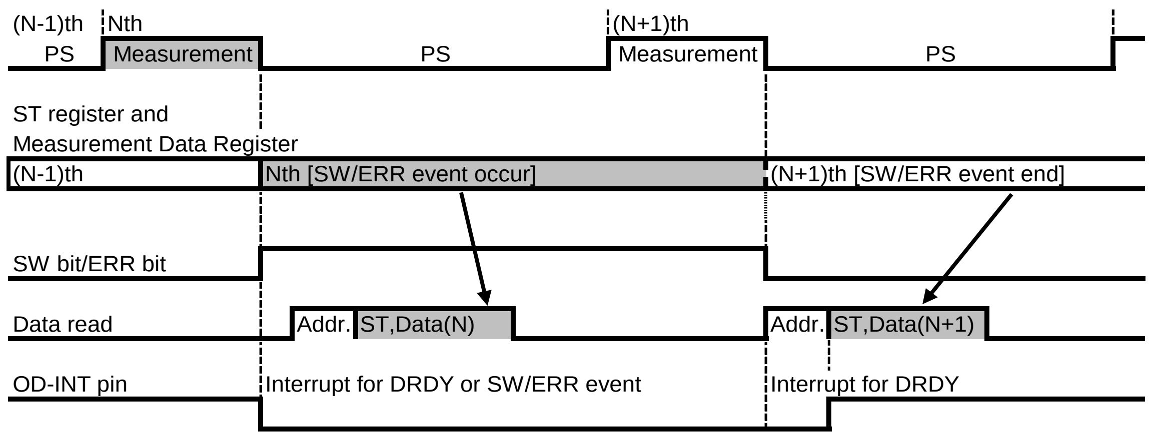

Figure 10.13 Timing chart of DRDY or SW/ERROR interrupt function

10.7.3. 当用户将OD-INT引脚分配给SW事件输出和/或错误时的开关/错误中断功能操作时序

事件输出，OD-INT引脚向用户通知这些事件。这些中断功能操作的时序如下所示。

表 10.5 软件/错误中断功能操作时序 (POL位 = “0”)

引脚名称	输出转换	转换时序
OD-INT引脚	H → L	测量结束 测量(软件/错误)
	L → H	测量结束 (软件/错误) or 写入地址 20h - 25h

图 10.12 软件/错误中断功能时序图

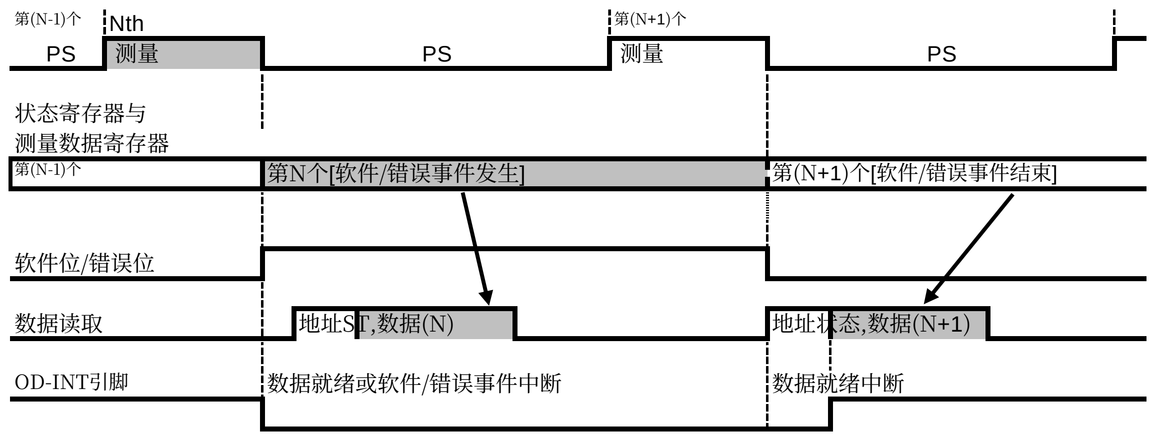

图 10.13 数据就绪或软件/错误中断功能时序图

10.8. Sensor Drive Select

Users can choose “Low power” or “Low noise” drive by the SDR bit.
“Low power” is used to save the current consumption and “Low noise” is used to reduce the noise of AK09973D. When Low noise (SDR bit = “0”) is set, output magnetic data noise is more reduced than Low power (about 70% of Low power). When Low power (SDR bit = “1”) is set, average current consumption at 10 Hz repetition rate is saved from 11 μA to 3.5 μA (Vdd = 1.8 V, +25˚C). Default SDR bit is Low noise drive (SDR bit = “0”).

10.9. Sensor Measurement Range and Sensitivity Select

Users can choose “High sensitivity (Normal measurement range and high sensitivity)” or “Wide range (Wide measurement range and normal sensitivity)” setting.
“High sensitivity” is used to measure with high magnetic sensitivity and “Wide range” is used to measure strong magnetic field (apply only to Z-axis). When High sensitivity (SMR bit = “0”) is set, magnetic sensor sensitivity is about three times higher than Wide range (3.1 μT/LSB → 1.1 μT/LSB). When Wide range (SMR bit = “1”) is set, Z-axis measurement range is about three times wider than High sensitivity (Z-axis measurement range: ±36.04 mT → ±101.57 mT). Default SMR bit is High sensitivity enable (SMR bit = “0”).

10.8. 传感器驱动选择用户

可通过SDR位选择“低功耗”或“低噪声”驱动。

“低功耗”用于节省电流消耗，“低噪声”用于降低AK09973D的噪声。当设置低噪声（SDR 位 = “0”）时，输出磁数据噪声比低功耗模式（约为低功耗模式的70%）进一步降低。当设置低功耗（SDR位 = “1”）时，在10赫兹重复频率下的平均电流消耗可从11微安节省至3.5微安（Vdd = 1.8 V，+25℃）。默认SDR位为低噪声驱动（SDR位 = “0”）。

10.9. 传感器测量范围与灵敏度选择用户

可选择“高灵敏度（正常测量范围与高灵敏度）”或“宽范围（宽测量范围与正常灵敏度）”设置。

“高灵敏度”用于以高磁灵敏度进行测量，而“宽范围”用于测量强磁场（仅适用于Z轴）。当设置高灵敏度（SMR位= “0”）时，磁传感器灵敏度比宽范围模式高约三倍（3.1 μT/LSB → 1.1 μT/LSB）。当设置宽范围（SMR位= “1”）时，Z轴测量范围比高灵敏度模式宽约三倍（Z轴测量范围：±36.04 mT → ±101.57 mT）。默认SMR位为启用高灵敏度（SMR位= “0”）。

11. Serial Interface

11.1. I^{2}C Bus Interface

The I^{2}C bus interface of AK09973D supports the Standard mode (100 kHz max.), the Fast mode (400 kHz max.) and the Fast mode plus (1000 kHz max.).

11.1.1. Data Transfer

To access AK09973D on the bus, generate a start condition first.
Next, transmit a one-byte slave address including a device address. At this time, AK09973D compares the slave address with its own address. If these addresses match, AK09973D generates an acknowledgement, and then executes READ or WRITE instruction. At the end of instruction execution, generate a stop condition.

11.1.1.1. Change of Data

A change of data on the SDA line must be made during "Low" period of the clock on the SCL line. When the clock signal on the SCL line is "High", the state of the SDA line must be stable. (Data on the SDA line can be changed only when the clock signal on the SCL line is "Low".)
During the SCL line is "High", the state of data on the SDA line is changed only when a start condition or a stop condition is generated.

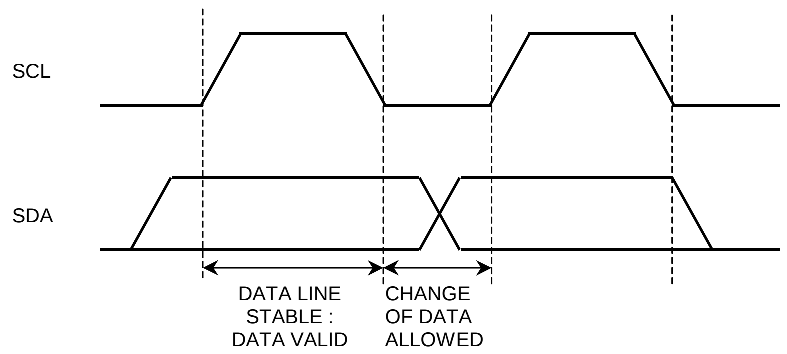

Figure 11.1 Data Change

11.1.1.2. Start/Stop Condition

If the SDA line is driven to "Low" from "High" when the SCL line is "High", a start condition is generated. Every instruction starts with a start condition.
If the SDA line is driven to "High" from "Low" when the SCL line is "High", a stop condition is generated. Every instruction stops with a stop condition.

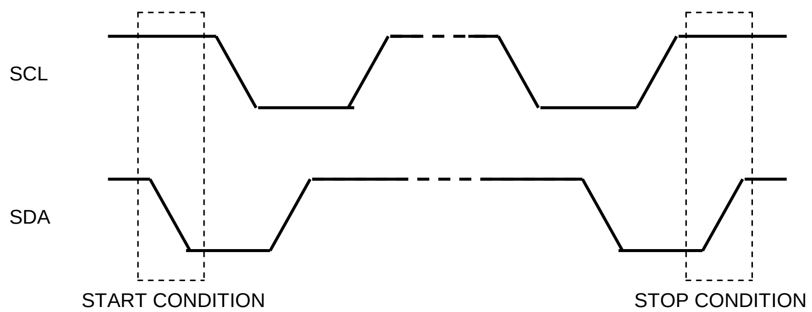

Figure 11.2 Start and stop condition

11. 串行接口

11.1. I^{2}C 总线接口AK09973D的I^{2}C总线接口

09973D支持标准模式（最大100千赫兹）、快速模式（最大400千赫兹）和增强快速模式（最大1000千赫兹）。

11.1.1. 数据传输

要访问总线上的AK09973D，首先产生一个起始条件。接着，发送一个包含设备地址的一字节从机地址。此时，AK09973D将该从机地址与自身地址进行比较。如果地址匹配，AK09973D将产生一个应答信号，然后执行读取或写入指令。指令执行结束时，产生一个停止条件。

11.1.1.1. 数据变化

SDA线上的数据变化必须在SCL线上的时钟信号为“低”电平期间进行。当SCL线上的时钟信号为“高”电平时，SDA线的状态必须保持稳定。（只有当SCL线上的时钟信号为“低”电平时，SDA线上的数据才能改变。）

在SCL线为“高电平”期间，SDA线上的数据状态仅在产生起始条件或停止条件时才会改变。

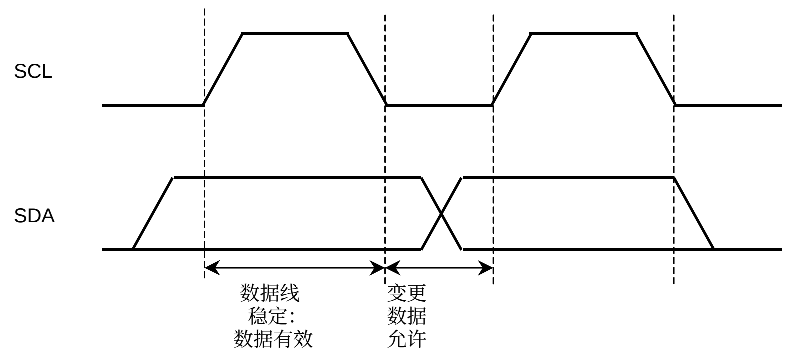

图 11.1 数据变化

11.1.1.2. 起始/停止条件

当SCL线为“高电平”时，若SDA线从“高电平”被驱动至“低电平”，则产生一个起始条件。每条指令均以起始条件开始。

当SCL线为“高”电平时，如果SDA线从“低”电平被驱动为“高”电平，则会产生一个停止条件。每条指令都以一个停止条件结束。

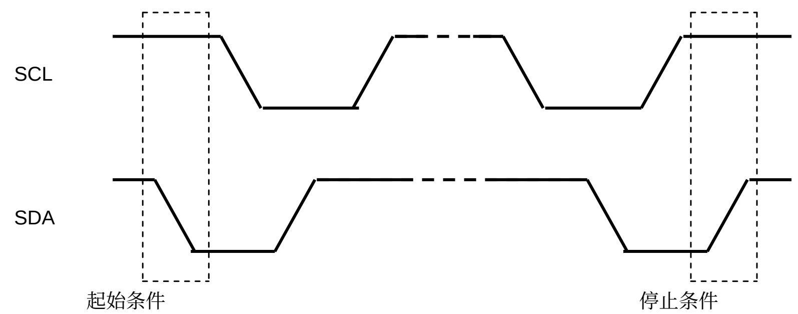

图 11.2 起始和停止条件

11.1.1.3. Acknowledge

The IC that is transmitting data releases the SDA line (in the "High" state) after sending 1-byte data. The IC that receives the data drives the SDA line to "Low" on the next clock pulse. This operation is referred as an acknowledge. With this operation, whether data has been transferred successfully can be checked. AK09973D generates an acknowledge after receipt of the start condition and slave address.
When a WRITE instruction is executed, AK09973D generates an acknowledge after every byte that is received.
When a READ instruction is executed, AK09973D generates an acknowledge then transfers the data stored at the specified address. Next, AK09973D releases the SDA line then monitors the SDA line. If a master IC generates an acknowledge instead of a stop condition, AK09973D transmits the 8-bit data stored at the next address. If no acknowledge is generated, AK09973D stops data transmission.

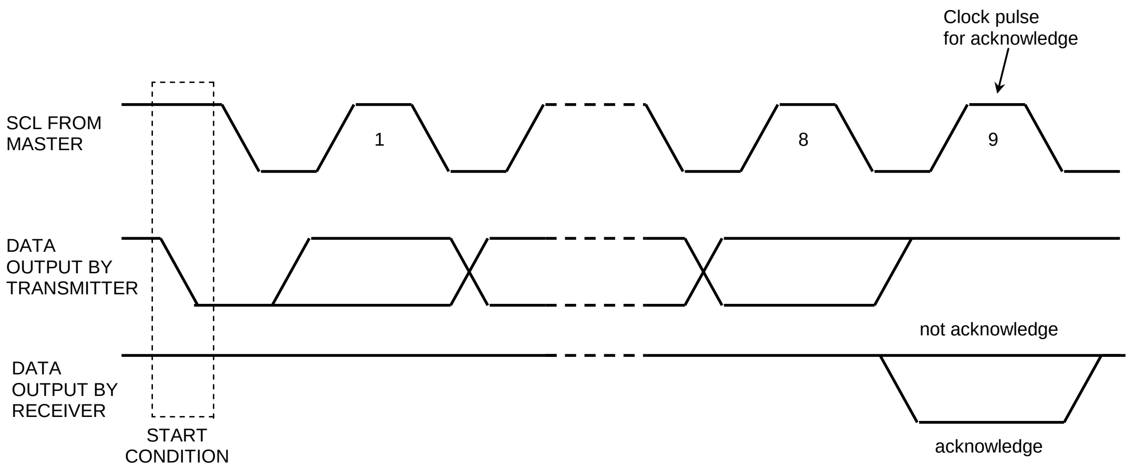

Figure 11.3 Generation of acknowledge

11.1.1.4. Slave Address

The slave address of AK09973D can be selected from the following list by changing pin connections of IF1 and IF2 pin.

	IF1	IF2	Slave address
Connection 1	OD-INT	SDA	10h
Connection 2	SDA	OD-INT	11h

MSB — LSB

0	0	1	0	0	0	0	R/W

Figure 11.4 Slave address of Connection 1

The first byte including a slave address is transmitted after a start condition, and an IC to be accessed is selected from the ICs on the bus according to the slave address.
When a slave address is transferred, the IC whose device address matches the transferred slave address generates an acknowledge then executes an instruction. The 8th bit (least significant bit) of the first byte is a R/W bit.
When the R/W bit is set to "1", READ instruction is executed. When the R/W bit is set to "0", WRITE instruction is executed.

11.1.1.3. 应答信号

发送数据的集成电路在发送完1字节数据后，会释放SDA线（使其处于“高”电平状态）。接收数据的集成电路会在下一个时钟脉冲将SDA线驱动为“低”电平。此操作被称为应答信号。通过此操作，可以检查数据是否已成功传输。AK09973D在接收到起始条件和从机地址后会产生一个应答信号。

当执行写入指令时，AK09973D 在接收到每个字节后都会生成一个应答信号。

当执行读取指令时，AK09973D会生成一个应答信号，然后传输存储在指定地址的数据。接着，AK09973D释放SDA线并监控SDA线。如果主控集成电路生成一个应答信号而非停止条件，AK09973D将传输存储在下个地址的8位数据。如果未生成应答信号，AK09973D则停止数据传输。

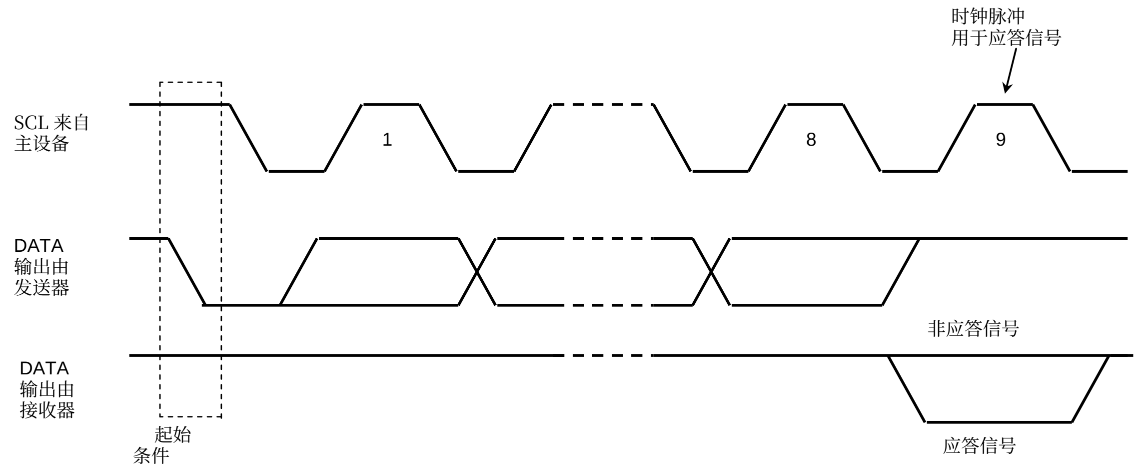

图11.3 应答信号的生成

11.1.1.4. 从机地址 AK0

9973D的从机地址可通过改变IF1和IF2引脚的连接方式，从以下列表中选择。

	IF1	IF2	从机地址
连接1	OD-INT	SDA	10h
连接2	SDA	OD-INT	11h

MSB — LSB

0	0	1	0	0	0	0	R/W

图 11.4 连接1的从机地址

包含从机地址的第一个字节在起始条件之后传输，根据该从机地址，从总线上的集成电路中选择要访问的集成电路。

当从机地址被传输时，设备地址与所传输从机地址匹配的集成电路会产生一个应答信号，然后执行一条指令。第一个字节的第8位（最低有效位）是一个读/写位。

当读/写位设置为“1”时，执行读取指令。当读/写位设置为“0”时，执行写入指令。

11.1.2. WRITE Instruction

When the R/W bit is set to "0", AK09973D performs write operation.
In write operation, AK09973D generates an acknowledge after receiving a start condition and the first byte (slave address) then receives the second byte. The second byte is used to specify the address of an internal control register and is based on the MSB-first configuration.

MSB							LSB
A7	A6	A5	A4	A3	A2	A1	A0

Figure 11.5 Register address

After receiving the second byte (register address), AK09973D generates an acknowledge then receives the third byte.
The third and the following bytes represent control data. Control data consists of 8-bit and is based on the MSB-first configuration. AK09973D generates an acknowledge after every byte is received. Data transfer always stops with a stop condition generated by the master.

MSB							LSB
D7	D6	D5	D4	D3	D2	D1	D0

Figure 11.6 Control data

AK09973D can write multiple bytes of data at a time.
After reception of the third byte (control data), AK09973D generates an acknowledge then receives the next data. If additional data is received instead of a stop condition after receiving one byte of data, the address counter inside the LSI chip is automatically incremented and the data is written at the next address.
The address is incremented from 20h to 25h. When the address is between 20h and 25h, the address is incremented 20h → 21h → 22h → 23h → 24h → 25h, and the address goes back to 20h after 25h. Actual data is written only to Read/Write registers (Table 12.2).

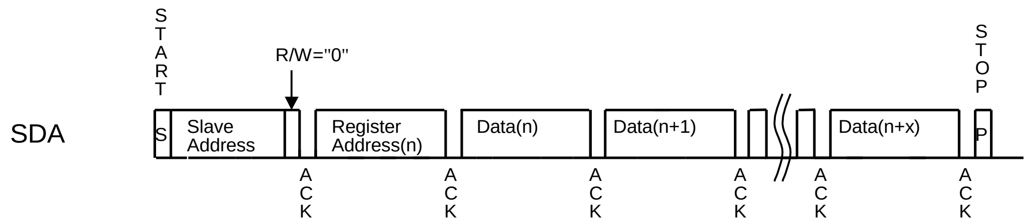

Figure 11.7 WRITE Instruction

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11.1.2. 写入指令

当读/写位设置为"0"时，AK09973D执行写入操作。
在写入操作中，AK09973D 在接收到起始条件和第一个字节（从机地址）后生成一个应答信号，随后接收第二个字节。第二个字节用于指定一个内部控制寄存器的地址，并基于最高有效位优先配置。

MSB							LSB
A7	A6	A5	A4	A3	A2	A1	A0

图 11.5 寄存器地址

接收到第二个字节（寄存器地址）后，AK09973D 生成一个应答信号，随后接收第三个字节。

第三个及后续字节代表控制数据。控制数据由 8 位组成，并基于最高有效位优先配置。AK09973D 在接收到每个字节后都会生成一个应答信号。数据传输总是在主设备生成的停止条件处结束。

MSB							LSB
D7	D6	D5	D4	D3	D2	D1	D0

图 11.6 控制数据

AK09973D 可以一次写入多个字节的数据。
接收到第三个字节（控制数据）后，AK09973D 会生成一个应答信号，然后接收下一个数据。如果在接收一个字节的数据后，接收到的是额外数据而非停止条件，则大规模集成电路芯片内部的地址计数器会自动递增，数据将被写入下一个地址。
地址从20h递增至25h。当地址在20h到25h之间时，地址按20h → 21h → 22h → 23h → 24h → 25 h递增，并在25h之后回到20h。实际数据仅写入读/写寄存器（表 12.2）。

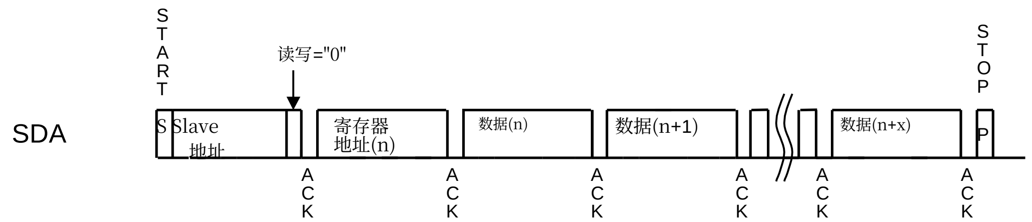

图 11.7 写入指令

11.1.3. READ Instruction

When the R/W bit is set to “1”, AK09973D performs read operation.
If a master IC generates an acknowledge instead of a stop condition after AK09973D transfers the data at a specified address, the data at the next address can be read.
Address can be 20h to 25h. When the address is between 20h and 25h , the address is incremented 20h → 21h → 22h → 23h → 24h → 25h , and the address goes back to 20h after 25h .
AK09973D supports one byte read and multiple byte read.

11.1.3.1. Current Address Read

AK09973D has an address counter inside the LSI chip. In current address read operation, the data at an address specified by this counter is read.
The internal address counter holds the next address of the most recently accessed address.
For example, if the address most recently accessed (for READ instruction) is address “n”, and a current address read operation is attempted, the data at address “n+1” is read.
In current address read operation, AK09973D generates an acknowledge after receiving a slave address for the READ instruction (R/W bit = “1”). Next, AK09973D transfers the data specified by the internal address counter starting with the next clock pulse, then increments the internal counter by one. If the master IC generates a stop condition instead of an acknowledge after AK09973D transmits one byte of data, the read operation stops.

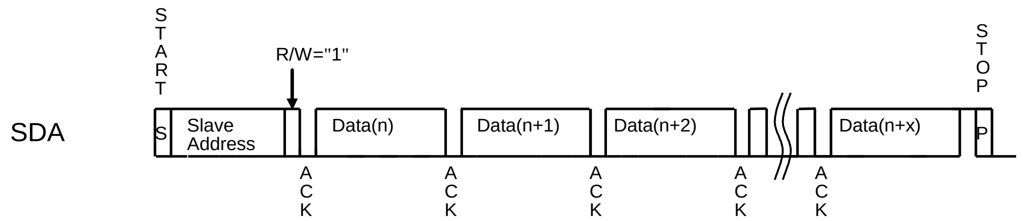

Figure 11.8 Current address read

11.1.3.2. Random Address Read

By random address read operation, data at an arbitrary address can be read.
The random address read operation requires to execute WRITE instruction as dummy before a slave address for the READ instruction (R/W bit = “1”) is transmitted. In random read operation, a start condition is first generated then a slave address for the WRITE instruction (R/W bit = “0”) and a read address are transmitted sequentially.
After AK09973D generates an acknowledge in response to this address transmission, a start condition and a slave address for the READ instruction (R/W bit = “1”) are generated again. AK09973D generates an acknowledge in response to this slave address transmission. Next, AK09973D transfers the data at the specified address then increments the internal address counter by one. If the master IC generates a stop condition instead of an acknowledge after data is transferred, the read operation stops.

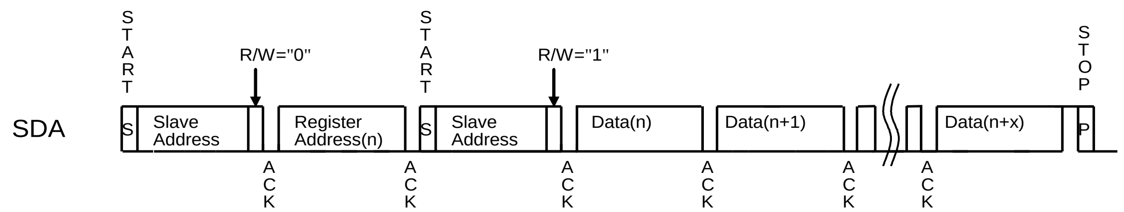

Figure 11.9 Random address read

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11.1.3. 当 th 时的读取指令

读/写位被设置为“1”，AK09973D 执行读取操作。
如果主控集成电路在AK09973D传输数据后，不是发出停止条件，而是发出应答信号， a 则可以读取下一个地址的数据。地址可以是20h到25h。当地址在20h到25h之间时，地址会按20h → 21h → 22h → 23h → 24h → 25h递增，并在25h之后回到20h。AK09973D支持单字节读取和多字节读取。

11.1.3.1. 当前地址读取 AK09973D内部有一个地址计数

器。在当前地址读取操作中，读取的是由该计数器指定的地址处的数据。
内部地址计数器保存着最近访问地址的下一个地址。
例如，如果最近访问的地址（用于读取指令）是地址“n”，并且尝试进行当前地址读取操作，那么读取的将是地址“n+1”处的数据。
在当前地址读取操作中，AK09973D 接收到针对读取指令的从机地址后生成一个应答信号（读/写位= 为“1”）。接着，AK09973D 从下一个时钟脉冲开始传输由内部地址计数器指定的数据，然后将内部计数器加一。如果主控集成电路在 AK09973D 传输一个字节的数据后生成停止条件而非应答信号，则读取操作停止。

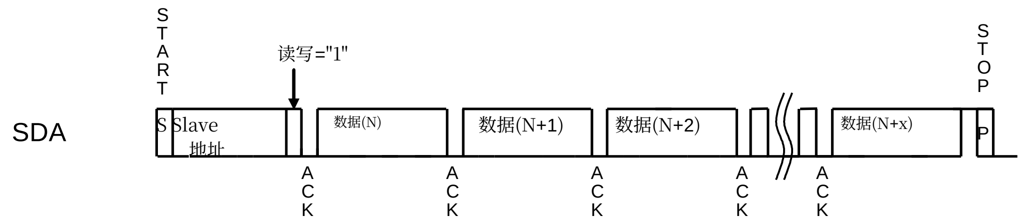

图 11.8 当前地址读取

11.1.3.2. 随机地址读取 通过随机地址读取操作， 可以读取任意地址的数据。
随机地址读取操作需要在传输针对读取指令的从机地址（读/写位 “1”）之前，先执行一次写入指令作为虚操作。在随机读取操作中，首先生成一个起始条件，然后依次传输针对写入指令的从机地址（读/写位 “0”）和一个读取地址。
在AK09973D针对此次地址传输生成应答信号后，会再次生成一个起始条件以及用于读取指令（R/W位为 = “1”）的从机地址。AK09973D会针对此次从机地址传输生成应答信号。接着，AK09973D传输指定地址处的数据，然后将内部地址计数器加一。如果在数据传输后，主控集成电路生成的是停止条件而非应答信号，则读取操作停止。

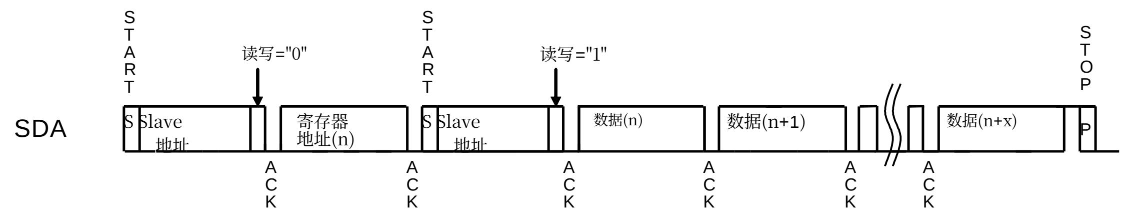

图 11.9 随机地址读取

12. Registers

12.1. Description of Registers

AK09973D has registers of 26 addresses as indicated in Table 12.1. Every address consists of 1-byte to 7-byte data. Data is transferred to or received from the external CPU via the serial interface described previously.

Table 12.1 Register Table

Address	READ/ WRITE	Description	Byte width	Remarks
00h	READ	Company ID, Device ID	4	Device Information
10h		Status	1	ST data
11h		Status and Measurement Magnetic Data	3	ST + X-axis data
12h			3	ST + Y-axis data
13h			5	ST + X and Y-axis data
14h			3	ST + Z-axis data
15h			5	ST + X and Z-axis data
16h			5	ST + Y and Z-axis data
17h			7	ST + X, Y and Z-axis data
18h			5	ST + Sum of squares of 3-axis data
19h		Status and Measurement Magnetic Data (upper 8 bits of measurement data register)	2	ST + X-axis data
1Ah			2	ST + Y-axis data
1Bh			3	ST + X and Y-axis data
1Ch			2	ST + Z-axis data
1Dh			3	ST + X and Z-axis data
1Eh			3	ST + Y and Z-axis data
1Fh			4	ST + X, Y and Z-axis data
20h	READ/ WRITE	Control 1	2	Interrupt function settings
21h		Control 2	1	Operation Mode, Sensor Drive, Measurement Range and Sensitivity
22h		Control 3 (Switch threshold value)	4	X-axis threshold settings
23h			4	Y-axis threshold settings
24h			4	Z-axis threshold settings
25h			4	Sum of squares of 3-axis threshold settings
30h		Reset	1	Soft reset
40h		Test	2	DO NOT ACCESS
41h	READ		1	DO NOT ACCESS

Addresses 20h to 25h are compliant with automatic increment function of serial interface respectively. When the address is in 20h to 25h , the address is incremented 20h → 21h → 22h → 23h → 24h → 25h, and the address goes back to 20h after 25h .

12. 寄存器

12.1. 寄存器描述AK09973D 拥有 26 个地址的寄存器

过程如 表12.1所示。每个地址由1字节至7字节的数据组成。数据通过前述串行接口传输至外部CPU或从外部CPU接收。

表12.1 寄存器表

地址	读取/写入	描述	Byte 宽度	备注
00h	READ	公司ID, 设备ID	4	设备信息
10h		状态	1	ST数据
11h		状态 and 测量磁数据	3	ST + X轴数据
12h			3	ST + Y轴数据
13h			5	ST + X轴和Y轴数据
14h			3	ST + Z轴数据
15h			5	ST + X轴和Z轴数据
16h			5	ST + Y轴和Z轴数据
17h			7	ST + X、Y和Z轴数据
18h			5	ST + 平方和 三轴数据
19h		状态 and 测量磁数据 (测量数据寄存器的高8位)	2	ST + X轴数据
1Ah			2	ST + Y轴数据
1Bh			3	ST + X轴和Y轴数据
1Ch			2	ST + Z轴数据
1Dh			3	ST + X轴和Z轴数据
1Eh			3	ST + Y轴和Z轴数据
1Fh			4	ST + X轴、Y轴和Z轴数据
20h	读取/写入	控制寄存器1	2	中断功能设置
21h		控制2	1	操作模式，传感器驱动，测量范围和灵敏度
22h		控制3 （开关阈值）	4	X轴阈值设置
23h			4	Y轴阈值设置
24h			4	Z轴阈值设置
25h			4	平方和 三轴阈值设置
30h		复位	1	软复位
40h		Test	2	禁止访问
41h	READ		1	禁止访问

地址20h至25h分别符合串行接口的自动递增功能。当地址在20h至25h范围内时，地址按20h → 21 h → 22h → 23h → 24h →25h的顺序递增，并在25h之后返回到20h。

12.2. Register Map

Table 12.2 Register Map

Addr.	Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6
Read only register							
00h	WIA[15:8]	WIA[7:0]	RSV[15:8]	RSV[7:0]	-	-	-
10h	ST[7:0]	-	-	-	-	-	-
11h	ST[7:0]	HX[15:8]	HX[7:0]	-	-	-	-
12h	ST[7:0]	HY[15:8]	HY[7:0]	-	-	-	-
13h	ST[7:0]	HY[15:8]	HY[7:0]	HX[15:8]	HX[7:0]	-	-
14h	ST[7:0]	HZ[15:8]	HZ[7:0]	-	-	-	-
15h	ST[7:0]	HZ[15:8]	HZ[7:0]	HX[15:8]	HX[7:0]	-	-
16h	ST[7:0]	HZ[15:8]	HZ[7:0]	HY[15:8]	HY[7:0]	-	-
17h	ST[7:0]	HZ[15:8]	HZ[7:0]	HY[15:8]	HY[7:0]	HX[15:8]	HX[7:0]
18h	ST[7:0]	HV[31:24]	HV[23:16]	HV[15:8]	HV[7:0]	-	-
19h	ST[7:0]	HX[15:8]	-	-	-	-	-
1Ah	ST[7:0]	HY[15:8]	-	-	-	-	-
1Bh	ST[7:0]	HY[15:8]	HX[15:8]	-	-	-	-
1Ch	ST[7:0]	HZ[15:8]	-	-	-	-	-
1Dh	ST[7:0]	HZ[15:8]	HX[15:8]	-	-	-	-
1Eh	ST[7:0]	HZ[15:8]	HY[15:8]	-	-	-	-
1Fh	ST[7:0]	HZ[15:8]	HY[15:8]	HX[15:8]	-	-	-
Read/Write register							
20h	CNTL1[15:8]	CNTL1[7:0]		-	-	-	-
21h	CNTL2[7:0]	-	-	-	-	-	-
22h	BOPX[15:8]	BOPX[7:0]	BRPX[15:8]	BRPX[7:0]	-	-	-
23h	BOPY[15:8]	BOPY[7:0]	BRPY[15:8]	BRPY[7:0]	-	-	-
24h	BOPZ[15:8]	BOPZ[7:0]	BRPZ[15:8]	BRPZ[7:0]	-	-	-
25h	BOPV[15:8]	BOPV[7:0]	BRPV[15:8]	BRPV[7:0]	-	-	-
30h	SRST[7:0]	-	-	-	-	-	-
40h	TEST1[15:8]	TEST1[7:0]	-	-	-	-	-
41h	TEST2[7:0]	-	-	-	-	-	-

12.2. 寄存器映射表

表 12.2 寄存器映射表

地址	字节0	字节1	字节2	字节3	字节4	字节5	字节6
只读寄存器							
00h	公司及设备ID寄存器[15:8]	公司及设备ID寄存器[7:0]	保留位[15:8]	保留位[7:0]	-	-	-
10h	状态[7:0]	-	-	-	-	-	-
11h	状态[7:0]	HX寄存器[15:8]	HX寄存器[7:0]	-	-	-	-
12h	状态[7:0]	HY寄存器[15:8]	HY寄存器[7:0]	-	-	-	-
13h	状态[7:0]	HY寄存器[15:8]	HY寄存器[7:0]	HX寄存器[15:8]	HX寄存器[7:0]	-	-
14h	状态[7:0]	赫兹[15:8]	赫兹[7:0]	-	-	-	-
15h	状态[7:0]	赫兹[15:8]	赫兹[7:0]	HX寄存器[15:8]	HX寄存器[7:0]	-	-
16h	状态[7:0]	赫兹[15:8]	赫兹[7:0]	HY寄存器[15:8]	HY寄存器[7:0]	-	-
17h	状态[7:0]	赫兹[15:8]	赫兹[7:0]	HY寄存器[15:8]	HY寄存器[7:0]	HX寄存器[15:8]	HX寄存器[7:0]
18h	状态[7:0]	HV寄存器[31:24]	HV寄存器[23:16]	HV寄存器[15:8]	HV寄存器[7:0]	-	-
19h	状态[7:0]	HX寄存器[15:8]	-	-	-	-	-
1Ah	状态[7:0]	HY寄存器[15:8]	-	-	-	-	-
1Bh	状态[7:0]	HY寄存器[15:8]	HX寄存器[15:8]	-	-	-	-
1Ch	状态[7:0]	赫兹[15:8]	-	-	-	-	-
1Dh	状态[7:0]	赫兹[15:8]	HX寄存器[15:8]	-	-	-	-
1Eh	状态[7:0]	赫兹[15:8]	HY寄存器[15:8]	-	-	-	-
1Fh	状态[7:0]	赫兹[15:8]	HY寄存器[15:8]	HX寄存器[15:8]	-	-	-
读/写寄存器							
20h	CNTL1[15:8]	CNTL1[7:0]		-	-	-	-
21h	CNTL2[7:0]	-	-	-	-	-	-
22h	BOPX[15:8]	BOPX[7:0]	BRPX[15:8]	BRPX[7:0]	-	-	-
23h	BOPY[15:8]	BOPY[7:0]	BRPY[15:8]	BRPY[7:0]	-	-	-
24h	BOPZ[15:8]	BOPZ[7:0]	BRPZ[15:8]	BRPZ[7:0]	-	-	-
25h	BOPV[15:8]	BOPV[7:0]	BRPV[15:8]	BRPV[7:0]	-	-	-
30h	SRST[7:0]	-	-	-	-	-	-
40h	测试1[15:8]	测试1[7:0]	-	-	-	-	-
41h	测试2[7:0]	-	-	-	-	-	-

Table 12.3 Further details about Register Map (D[7:0])

Register name	Bit number (D[7:0])							
	7	6	5	4	3	2	1	0
WIA[7:0]	1	1	0	0	0	0	0	1
RSV[7:0]	RSV7	RSV6	RSV5	RSV4	RSV3	RSV2	RSV1	RSV0
ST[7:0]	1	DOR	ERR	SWV	SWZ	SWY	SWX	DRDY
HX[7:0]	HX7	HX6	HX5	HX4	HX3	HX2	HX1	HX0
HY[7:0]	HY7	HY6	HY5	HY4	HY3	HY2	HY1	HY0
HZ[7:0]	HZ7	HZ6	HZ5	HZ4	HZ3	HZ2	HZ1	HZ0
HV[7:0]	HV7	HV6	HV5	HV4	HV3	HV2	HV1	HV0
CNTL1[7:0]	0	0	ERROREN	SWVEN	SWZEN	SWYEN	SWXEN	DRDYEN
CNTL2[7:0]	SELFT	SMR	SDR	MODE4	MODE3	MODE2	MODE1	MODE0
BOPX[7:0]	BOPX7	BOPX6	BOPX5	BOPX4	BOPX3	BOPX2	BOPX1	BOPX0
BRPX[7:0]	BRPX7	BRPX6	BRPX5	BRPX4	BRPX3	BRPX2	BRPX1	BRPX0
BOPY[7:0]	BOPY7	BOPY6	BOPY5	BOPY4	BOPY3	BOPY2	BOPY1	BOPY0
BRPY[7:0]	BRPY7	BRPY6	BRPY5	BRPY4	BRPY3	BRPY2	BRPY1	BRPY0
BOPZ[7:0]	BOPZ7	BOPZ6	BOPZ5	BOPZ4	BOPZ3	BOPZ2	BOPZ1	BOPZ0
BRPZ[7:0]	BRPZ7	BRPZ6	BRPZ5	BRPZ4	BRPZ3	BRPZ2	BRPZ1	BRPZ0
BOPV[7:0]	BOPV7	BOPV6	BOPV5	BOPV4	BOPV3	BOPV2	BOPV1	BOPV0
BRPV[7:0]	BRPV7	BRPV6	BRPV5	BRPV4	BRPV3	BRPV2	BRPV1	BRPV0
SRST[7:0]	0	0	0	0	0	0	0	SRST
TEST1[7:0]	-	-	-	-	-	-	-	-
TEST2[7:0]	-	-	-	-	-	-	-	-

Table 12.4 Further details about Register Map (D[15:8])

Register name	Bit number (D[15:8])							
	15	14	13	12	11	10	9	8
WIA[15:8]	0	1	0	0	1	0	0	0
RSV[15:8]	RSV15	RSV14	RSV13	RSV12	RSV11	RSV10	RSV9	RSV8
HX[15:8]	HX15	HX14	HX13	HX12	HX11	HX10	HX9	HX8
HY[15:8]	HY15	HY14	HY13	HY12	HY11	HY10	HY9	HY8
HZ[15:8]	HZ15	HZ14	HZ13	HZ12	HZ11	HZ10	HZ9	HZ8
HV[15:8]	HV15	HV14	HV13	HV12	HV11	HV10	HV9	HV8
CNTL1[15:8]	0	0	0	0	POLV	POLZ	POLY	POLX
BOPX[15:8]	BOPX15	BOPX14	BOPX13	BOPX12	BOPX11	BOPX10	BOPX9	BOPX8
BRPX[15:8]	BRPX15	BRPX14	BRPX13	BRPX12	BRPX11	BRPX10	BRPX9	BRPX8
BOPY[15:8]	BOPY15	BOPY14	BOPY13	BOPY12	BOPY11	BOPY10	BOPY9	BOPY8
BRPY[15:8]	BRPY15	BRPY14	BRPY13	BRPY12	BRPY11	BRPY10	BRPY9	BRPY8
BOPZ[15:8]	BOPZ15	BOPZ14	BOPZ13	BOPZ12	BOPZ11	BOPZ10	BOPZ9	BOPZ8
BRPZ[15:8]	BRPZ15	BRPZ14	BRPZ13	BRPZ12	BRPZ11	BRPZ10	BRPZ9	BRPZ8
BOPV[15:8]	BOPV15	BOPV14	BOPV13	BOPV12	BOPV11	BOPV10	BOPV9	BOPV8
BRPV[15:8]	BRPV15	BRPV14	BRPV13	BRPV12	BRPV11	BRPV10	BRPV9	BRPV8
TEST1[15:8]	-	-	-	-	-	-	-	-

Table 12.5 Further details about Register Map (D[23:16])

Register name	Bit number (D[23:16])							
	23	22	21	20	19	18	17	16
HV[23:16]	HV23	HV22	HV21	HV20	HV19	HV18	HV17	HV16

Table 12.6 Further details about Register Map (D[31:24])

Register name	Bit number (D[31:24])							
	31	30	29	28	27	26	25	24
HV[31:24]	HV31	HV30	HV29	HV28	HV27	HV26	HV25	HV24

TEST1 and TEST2 are test registers for shipment test. Do not access these registers.

表 12.3 寄存器映射表详情（D[7:0]）

寄存器 name	位号 (D[7:0])							
	7	6	5	4	3	2	1	0
公司及设备ID寄存器[7:0]	1	1	0	0	0	0	0	1
保留位[7:0]	RSV7	RSV6	RSV5	RSV4	RSV3	RSV2	RSV1	RSV0
状态[7:0]	1	DOR	ERR	SWV	SWZ	SWY	SWX	DRDY
HX寄存器[7:0]	HX7	HX6	HX5	HX4	HX3	HX2	HX1	HX0
HY寄存器[7:0]	HY7	HY6	HY5	HY4	HY3	HY2	HY1	HY0
赫兹[7:0]	HZ7	HZ6	HZ5	HZ4	HZ3	HZ2	HZ1	HZ0
HV寄存器[7:0]	HV7	HV6	HV5	HV4	HV3	HV2	HV1	HV0
CNTL1[7:0]	0	0	ERROREN	SWVEN	SWZEN	SWYEN	SWXEN	DRDYEN
CNTL2[7:0]	SELFT	SMR	SDR	MODE4	MODE3	MODE2	MODE1	MODE0
BOPX[7:0]	BOPX7	BOPX6	BOPX5	BOPX4	BOPX3	BOPX2	BOPX1	BOPX0
BRPX[7:0]	BRPX7	BRPX6	BRPX5	BRPX4	BRPX3	BRPX2	BRPX1	BRPX0
BOPY[7:0]	BOPY7	BOPY6	BOPY5	BOPY4	BOPY3	BOPY2	BOPY1	BOPY0
BRPY[7:0]	BRPY7	BRPY6	BRPY5	BRPY4	BRPY3	BRPY2	BRPY1	BRPY0
BOPZ[7:0]	BOPZ7	BOPZ6	BOPZ5	BOPZ4	BOPZ3	BOPZ2	BOPZ1	BOPZ0
BRPZ[7:0]	BRPZ7	BRPZ6	BRPZ5	BRPZ4	BRPZ3	BRPZ2	BRPZ1	BRPZ0
BOPV[7:0]	BOPV7	BOPV6	BOPV5	BOPV4	BOPV3	BOPV2	BOPV1	BOPV0
BRPV[7:0]	BRPV7	BRPV6	BRPV5	BRPV4	BRPV3	BRPV2	BRPV1	BRPV0
SRST[7:0]	0	0	0	0	0	0	0	SRST
测试1[7:0]	-	-	-	-	-	-	-	-
测试2[7:0]	-	-	-	-	-	-	-	-

表 12.4 寄存器映射表 (D[15:8]) 的详细信息

寄存器 name	位号 (D[15:8])							
	15	14	13	12	11	10	9	8
公司及设备ID寄存器[15:8]	0	1	0	0	1	0	0	0
保留位[15:8]	RSV15	RSV14	RSV13	RSV12	RSV11	RSV10	RSV9	RSV8
HX寄存器[15:8]	HX15	HX14	HX13	HX12	HX11	HX10	HX9	HX8
HY寄存器[15:8]	HY15	HY14	HY13	HY12	HY11	HY10	HY9	HY8
HZ[15:8]	HZ15	HZ14	HZ13	HZ12	HZ11	HZ10	HZ9	HZ8
HV寄存器[15:8]	HV15	HV14	HV13	HV12	HV11	HV10	HV9	HV8
CNTL1[15:8]	0	0	0	0	POLV	POLZ	POLY	POLX
BOPX[15:8]	BOPX15	BOPX14	BOPX13	BOPX12	BOPX11	BOPX10	BOPX9	BOPX8
BRPX[15:8]	BRPX15	BRPX14	BRPX13	BRPX12	BRPX11	BRPX10	BRPX9	BRPX8
BOPY[15:8]	BOPY15	BOPY14	BOPY13	BOPY12	BOPY11	BOPY10	BOPY9	BOPY8
BRPY[15:8]	BRPY15	BRPY14	BRPY13	BRPY12	BRPY11	BRPY10	BRPY9	BRPY8
BOPZ[15:8]	BOPZ15	BOPZ14	BOPZ13	BOPZ12	BOPZ11	BOPZ10	BOPZ9	BOPZ8
BRPZ[15:8]	BRPZ15	BRPZ14	BRPZ13	BRPZ12	BRPZ11	BRPZ10	BRPZ9	BRPZ8
BOPV[15:8]	BOPV15	BOPV14	BOPV13	BOPV12	BOPV11	BOPV10	BOPV9	BOPV8
BRPV[15:8]	BRPV15	BRPV14	BRPV13	BRPV12	BRPV11	BRPV10	BRPV9	BRPV8
测试1[15:8]	-	-	-	-	-	-	-	-

表 12.5 寄存器映射表（D[23:16]）的更多详细信息

寄存器 name	位号（D[23:16]）							
	23	22	21	20	19	18	17	16
HV[23:16]	HV23	HV22	HV21	HV20	HV19	HV18	HV17	HV16

表 12.6 寄存器映射表（D[31:24]）的更多详细信息

寄存器 name	位号 (D[31:24])							
	31	30	29	28	27	26	25	24
HV[31:24]	HV31	HV30	HV29	HV28	HV27	HV26	HV25	HV24

TEST1和TEST2是用于出货测试的测试寄存器。禁止访问这些寄存器 .

12.3. Detailed Description of Registers
12.3.1 WIA[15:0]: Company ID and Device ID

Addr.	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
00h	WIA[7:0]	1	1	0	0	0	0	0	1
Addr.	**Register name**	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
Read-only register									
00h	WIA[15:8]	0	1	0	0	1	0	0	0

WIA[7:0] bits: Device ID of AK09973D. It is described in one byte and fixed value.
C1h: fixed
WIA[15:8] bits: Company ID of AKM. It is described in one byte and fixed value.
48h: fixed

12.3.2 RSV[15:0]: Reserved Register

Addr.	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
00h	RSV[7:0]	RSV7	RSV6	RSV5	RSV4	RSV3	RSV2	RSV1	RSV0
Addr.	**Register name**	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
Read-only register									
00h	RSV[15:8]	RSV15	RSV14	RSV13	RSV12	RSV11	RSV10	RSV9	RSV8

RSV[7:0] bits/ RSV[15:8] bits: Reserved register for AKM.

12.3.3 ST[7:0]: Status

Addr.	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
10h-1fh	ST[7:0]	1	DOR	ERR	SWV	SWZ	SWY	SWX	DRDY
Reset		1	0	0	0	0	0	0	0

DRDY bit: Data Ready
“0”: Normal
“1”: Data is ready

DRDY bit turns to “1” when data is ready in Single measurement mode and Continuous measurement mode. It returns to “0” when any one of measurement data register (HX, HY, HZ or/and HV register) is read all the way through or access to Setting Registers (address 20h to 25h).

DOR bit: Data Overrun
“0”: Normal
“1”: Data overrun
DOR bit turns to “1” when data has been skipped in Continuous measurement mode. DOR bit turns to “0” at the after both of reading measurement data and the next measurement ended.

SWX bit, SWY bit, SWZ bit, SWV bit
“0”: Measurement data of X, Y, Z-axis and vector sum of 3-axis data is lower than returning threshold
“1”: Measurement data of X, Y, Z-axis and vector sum of 3-axis data is higher than operating threshold

12.3. 寄存器详细描述 12.3.1 WIA**[15:0]**: 公司ID和设备
ID

地址	寄存器 **name**	D7	D6	D5	D4	D3	D2	D1	D0
只读寄存器									
00h	公司及设备ID寄存器[7:0]	1	1	0	0	0	0	0	1
地址	寄存器 **name**	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
只读寄存器									
00h	WIA[15:8]	0	1	0	0	1	0	0	0

WIA[7:0] 位：AK09973D的设备ID。它用一个字节描述，为固定值。C1h：固定值。WIA[15:8] 位：AKM的公司ID。它用一个字节描述，为固定值。48h：固定值。

12.3.2 RSV**[15:0]**：保留寄存器

地址	寄存器 **name**	D7	D6	D5	D4	D3	D2	D1	D0
只读寄存器									
00h	RSV[7:0]	RSV7	RSV6	RSV5	RSV4	RSV3	RSV2	RSV1	RSV0
地址	寄存器 **name**	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
只读寄存器									
00h	RSV[15:8]	RSV15	RSV14	RSV13	RSV12	RSV11	RSV10	RSV9	RSV8

RSV[7:0] 位/ RSV[15:8] 位：为AKM保留的寄存器。

12.3.3 ST**[7:0]**：状态

地址	寄存器 **name**	D7	D6	D5	D4	D3	D2	D1	D0
只读寄存器									
10h-1fh	ST[7:0]	1	DOR	ERR	SWV	SWZ	SWY	SWX	DRDY
复位		1	0	0	0	0	0	0	0

DRDY位：数据就绪“0”：
正常 “1”：数据已就绪

在单次测量模式和连续测量模式下，当数据准备就绪时，DRDY位会变为“1”。当测量数据寄存器（HX、HY、HZ 和/或 HV 寄存器）中的任何一个被完全读取，或访问了设置寄存器（地址20h至25h）时，它会恢复为“0”。

数据溢出位：数据超限“0”：正常 “1”：数据超限在连续测量模式下，若数据被跳过，数据溢出位将变为“1”。数据溢出位将在读取测量数据且下一次测量结束后变为“0”。

SWX位、SWY位、SWZ位、SWV位
“0”：X、Y、Z轴测量数据及三轴数据矢量和低于返回阈值“1”：X、Y、Z轴测量数据及三轴数据矢量和高于操作阈值

ERR bit: Magnetic sensor overflow
“0”: Normal
“1”: Magnetic sensor overflow occurred

12.3.4 HX[15:0]/HY[15:0]/HZ[15:0]: Measurement Data

Addr.	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
11h	HX[7:0]	HX7	HX6	HX5	HX4	HX3	HX2	HX1	HX0
|	HY[7:0]	HY7	HY6	HY5	HY4	HY3	HY2	HY1	HY0
1fh	HZ[7:0]	HZ7	HZ6	HZ5	HZ4	HZ3	HZ2	HZ1	HZ0
Reset		0	0	0	0	0	0	0	0
Addr.	**Register name**	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
Read-only register									
11h	HX[15:8]	HX15	HX14	HX13	HX12	HX11	HX10	HX9	HX8
|	HY[15:8]	HY15	HY14	HY13	HY12	HY11	HY10	HY9	HY8
1fh	HZ[15:8]	HZ15	HZ14	HZ13	HZ12	HZ11	HZ10	HZ9	HZ8
Reset		0	0	0	0	0	0	0	0

Measurement data of magnetic sensor X-axis/Y-axis/Z-axis
HX[7:0] bits: X-axis measurement data lower 8-bit
HX[15:8] bits: X-axis measurement data higher 8-bit
HY[7:0] bits: Y-axis measurement data lower 8-bit
HY[15:8] bits: Y-axis measurement data higher 8-bit
HZ[7:0] bits: Z-axis measurement data lower 8-bit
HZ[15:8] bits: Z-axis measurement data higher 8-bit

Measurement data is stored in two’s complement. Measurement range of each axis is -32768 to 32767 in 16-bit output (High sensitivity setting). Measurement range of X and Y-axis are -11264 to 11264 in 16-bit output, Z-axis is -32768 to 32767 in 16-bit output (Wide range setting).

Table 12.7 Measurement magnetic data format (High sensitivity setting)

Measurement data (each axis) [15:0] bits			Magnetic flux density [mT]	ERR bit
Two’s complement	**Hex**	**Decimal**		
0111 1111 1111 1111	7FFF	32767	>36.0437	1
0111 1111 1111 1111	7FFF	32767	36.0437	0
|	|	|	|	|
0000 0000 0000 0001	0001	1	0.0011	0
0000 0000 0000 0000	0000	0	0	0
1111 1111 1111 1111	FFFF	-1	-0.0011	0
|	|	|	|	|
1000 0000 0000 0000	8000	-32768	-36.0448	0
1000 0000 0000 0000	8000	-32768	<-36.0448	1

错误位：磁传感器溢出“0”：正常“1”：发生磁传感器溢出

12.3.4 HX[15:0]/HY[15:0]/HZ[15:0]: 测量数据

地址	寄存器 name	D7	D6	D5	D4	D3	D2	D1	D0
只读寄存器									
11h	HX寄存器[7:0]	HX7	HX6	HX5	HX4	HX3	HX2	HX1	HX0
|	HY寄存器[7:0]	HY7	HY6	HY5	HY4	HY3	HY2	HY1	HY0
1fh	HZ寄存器[7:0]	HZ7	HZ6	HZ5	HZ4	HZ3	HZ2	HZ1	HZ0
复位		0	0	0	0	0	0	0	0
地址	寄存器 **name**	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
只读寄存器									
11h	HX寄存器[15:8]	HX15	HX14	HX13	HX12	HX11	HX10	HX9	HX8
|	HY寄存器[15:8]	HY15	HY14	HY13	HY12	HY11	HY10	HY9	HY8
1fh	HZ寄存器[15:8]	HZ15	HZ14	HZ13	HZ12	HZ11	HZ10	HZ9	HZ8
复位		0	0	0	0	0	0	0	0

磁传感器X轴/Y轴/Z轴测量数据 HX[7:0] 位：X轴测量数据低8位 HX[15:8] 位：X轴测量数据高8位 HY[7:0] 位：Y轴测量数据低8位 HY[15:8] 位：Y轴测量数据高8位 HZ[7:0] 位：Z轴测量数据低8位 HZ[15:8] 位：Z轴测量数据高8位

测量数据以二进制补码形式存储。各轴的测量范围在16位输出（高灵敏度设置）下为-32768至32767。在16位输出（宽范围设置）下，X轴和Y轴的测量范围为-11264至11264，Z轴为-32768至32767。

表12.7 测量磁数据格式（高灵敏度设置）

测量数据（各轴） **[15:0]** 位			磁通密度 [毫特斯拉]	错误位
二进制补码	**Hex**	十进制		
0111 1111 1111 1111	7FFF	32767	>36.0437	1
0111 1111 1111 1111	7FFF	32767	36.0437	0
|	|	|	|	|
0000 0000 0000 0001	0001	1	0.0011	0
0000 0000 0000 0000	0000	0	0	0
1111 1111 1111 1111	FFFF	-1	-0.0011	0
|	|	|	|	|
1000 0000 0000 0000	8000	-32768	-36.0448	0
1000 0000 0000 0000	8000	-32768	<-36.0448	1

Table 12.8 Measurement magnetic data format (Wide range setting, X and Y-axis)

Measurement data (X and Y axis) [15:0] bits			Magnetic flux density [mT]	ERR bit
Two's complement	Hex	Decimal		
0010 1100 0000 0000	2C00	11264	>34.9184	1
0010 1100 0000 0000	2C00	11264	34.9184	0
|	|	|	|	|
0000 0000 0000 0001	0001	1	0.0031	0
0000 0000 0000 0000	0000	0	0	0
1111 1111 1111 1111	FFFF	-1	-0.0031	0
|	|	|	|	|
1101 0100 0000 0000	D400	-11264	-34.9184	0
1101 0100 0000 0000	D400	-11264	<-34.9184	1

Table 12.9 Measurement magnetic data format (Wide range setting, Z-axis)

Measurement data (Z axis) [15:0] bits			Magnetic flux density [mT]	ERR bit
Two's complement	Hex	Decimal		
0111 1111 1111 1111	7FFF	32767	>101.5777	1
0111 1111 1111 1111	7FFF	32767	101.5777	0
|	|	|	|	|
0000 0000 0000 0001	0001	1	0.0031	0
0000 0000 0000 0000	0000	0	0	0
1111 1111 1111 1111	FFFF	-1	-0.0031	0
|	|	|	|	|
1000 0000 0000 0000	8000	-32768	-101.5808	0
1000 0000 0000 0000	8000	-32768	<-101.5808	1

12.3.5 HV[23:0]: Sum of Squares of 3-axis Measurement Data

Addr.	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read-only register									
18h	HV[7:0]	HX7	HX6	HX5	HX4	HX3	HX2	HX1	HX0
Reset		0	0	0	0	0	0	0	0
Addr.	**Register name**	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
Read-only register									
18h	HV[15:8]	HX15	HX14	HX13	HX12	HX11	HX10	HX9	HX8
Reset		0	0	0	0	0	0	0	0
Addr.	**Register name**	**D23**	**D22**	**D21**	**D20**	**D19**	**D18**	**D17**	**D16**
Read-only register									
18h	HV[23:16]	HX23	HX22	HX21	HX20	HX19	HX18	HX17	HX16
Reset		0	0	0	0	0	0	0	0
Addr.	**Register name**	**D31**	**D30**	**D29**	**D28**	**D27**	**D26**	**D25**	**D24**
18h	HV[31:24]	HX31	HX30	HX29	HX28	HX27	HX26	HX25	HX24
Reset		0	0	0	0	0	0	0	0

表12.8 测量磁数据格式（宽范围设置，X轴和Y轴）

测量数据（X轴和Y轴）[15:0] 位			磁通密度 [毫特斯拉]	错误位
二进制补码	Hex	十进制		
0010 1100 0000 0000	2C00	11264	>34.9184	1
0010 1100 0000 0000	2C00	11264	34.9184	0
|	|	|	|	|
0000 0000 0000 0001	0001	1	0.0031	0
0000 0000 0000 0000	0000	0	0	0
1111 1111 1111 1111	FFFF	-1	-0.0031	0
|	|	|	|	|
1101 0100 0000 0000	D400	-11264	-34.9184	0
1101 0100 0000 0000	D400	-11264	<-34.9184	1

表12.9 测量磁数据格式（宽范围设置，Z轴）

测量数据（Z轴） [15:0] 位			磁通密度 [毫特斯拉]	错误位
二进制补码	Hex	十进制		
0111 1111 1111 1111	7FFF	32767	>101.5777	1
0111 1111 1111 1111	7FFF	32767	101.5777	0
|	|	|	|	|
0000 0000 0000 0001	0001	1	0.0031	0
0000 0000 0000 0000	0000	0	0	0
1111 1111 1111 1111	FFFF	-1	-0.0031	0
|	|	|	|	|
1000 0000 0000 0000	8000	-32768	-101.5808	0
1000 0000 0000 0000	8000	-32768	<-101.5808	1

12.3.5 HV[23:0]: 三轴测量数据平方和

地址	寄存器 name	D7	D6	D5	D4	D3	D2	D1	D0
只读寄存器									
18h	HV[7:0]	HX7	HX6	HX5	HX4	HX3	HX2	HX1	HX0
复位		0	0	0	0	0	0	0	0
地址	寄存器 **name**	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
只读寄存器									
18h	HV[15:8]	HX15	HX14	HX13	HX12	HX11	HX10	HX9	HX8
复位		0	0	0	0	0	0	0	0
地址	寄存器 **name**	**D23**	**D22**	**D21**	**D20**	**D19**	**D18**	**D17**	**D16**
只读寄存器									
18h	HV[23:16]	HX23	HX22	HX21	HX20	HX19	HX18	HX17	HX16
复位		0	0	0	0	0	0	0	0
地址	寄存器 **name**	**D31**	**D30**	**D29**	**D28**	**D27**	**D26**	**D25**	**D24**
18h	HV[31:24]	HX31	HX30	HX29	HX28	HX27	HX26	HX25	HX24
复位		0	0	0	0	0	0	0	0

Sum of squares of 3-axis measurement data.
HV[31:0] = (HX[15:0])^2+ (HY[15:0])^2+ (HZ[15:0])^2
HV[7:0] bits: 3-axis measurement data lower 8-bit
HV[15:8] bits: 3-axis measurement data middle 8-bit
HV[23:16] bits: 3-axis measurement data middle 8-bit
HV[31:24] bits: 3-axis measurement data higher 8-bit

12.3.6 CNTL1[15:0]: Interrupt Output Setting

Addr.	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read/Write register									
20h	CNTL1[7:0]	0	0	ERREN	SWVEN	SWZEN	SWYEN	SWXEN	DRDYEN
Reset		0	0	0	0	0	0	0	0
Addr.	**Register name**	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
Read/Write register									
20h	CNTL1[15:8]	0	0	RSV* 16	0	POLV	POLZ	POLY	POLX
Reset		0	0	0	0	0	0	0	0

DRDYEN bit: DRDY event output
“0”: DRDY event outputs disable
“1”: DRDY event outputs enable

SWXEN bit to SWVEN bit: Switch event output
“0”: Switch event outputs disable
“1”: Switch event outputs enable

ERREN bit: ERR event output
“0”: ERR event outputs disable
“1”: ERR event outputs enable

POLX, POLY, POLZ and POLV bit: Polarity of interrupt event signal of OD-INT pin setting
“0”: Negative logic output
“1”: Positive logic output

Note:
* 16 Please write “0” on RSV bit when you write on address 20h.

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三轴测量数据平方和。HV[31:0] = (HX[15:0])^2+ (
HY[15:0])^2+ (HZ[15:0])^2HV[7:0] 位：三轴测量数据
低8位 HV[15:8] 位：三轴测量数据中8位 HV[23:16] 位：
三轴测量数据中8位 HV[31:24] 位：三轴测量数据高8位

12.3.6 CNTL1[15:0]：中断输出设置

地址	寄存器 name	D7	D6	D5	D4	D3	D2	D1	D0
读/写寄存器									
20h	CNTL1[7:0]	0	0	错误使能	SWVEN	SWZEN	SWYEN	SWXEN	数据就绪
复位		0	0	0	0	0	0	0	0
地址	寄存器 name	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
读/写寄存器									
20h	CNTL1[15:8]	0	0	保留位* 16	0	POLV	POLZ	POLY	POLX
复位		0	0	0	0	0	0	0	0

DRDYEN位：数据就绪事件输出“0”：数
据就绪事件输出禁用“1”：数据就绪事件输
出启用

SWXEN位至SWVEN位：切换事件输出“0”：
切换事件输出禁用“1”：切换事件输出启用

ERREN位：错误事件输出“0”：错
误事件输出禁用“1”：错误事件输出
启用

POLX、POLY、POLZ和POLV位：OD-INT引脚中断事件信号极性设置“0”：负逻辑输
出“1”：正逻辑输出

注意：* 16 请
在写入时，对RSV位写入“0” 在地址20h上。

12.3.7 CNTL2[7:0]: Operation Mode, Sensor Drive and Self-test Setting

Addr.	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read/Write register									
21h	CNTL2[7:0]	STEST	SMR	SDR	MODE4	MODE3	MODE2	MODE1	MODE0
Reset		0	0	0	0	0	0	0	0

MODE[4:0] bits: Operation mode setting
“01h”: Single measurement mode
“02h”: Continuous measurement mode 1
“04h”: Continuous measurement mode 2
“06h”: Continuous measurement mode 3
“08h”: Continuous measurement mode 4
“0Ah”: Continuous measurement mode 5
“0Ch”: Continuous measurement mode 6
“0Eh”: Continuous measurement mode 7
“10h”: Continuous measurement mode 8
“otherwise”: Power-down mode

SDR bit: Sensor drive setting
“0”: Low noise drive
“1”: Low power drive

SMR bit: Measurement range and sensitivity setting
“0”: High sensitivity setting
“1”: Wide measurement range setting

STEST bit: Self-test setting
“0”: Self-test disable
“1”: Self-test enable

12.3.7 CNTL2[7:0]: 操作模式、传感器驱动和自检设置

地址	寄存器 name	D7	D6	D5	D4	D3	D2	D1	D0
读/写寄存器									
21h	CNTL2[7:0]	自测试	SMR	SDR	模式4	模式3	模式2	模式1	模式0
复位		0	0	0	0	0	0	0	0

模式[4：0] 位：操作模式设置“01h”：
单次测量模式 “02h”：连续测量模式1
“04h”：连续测量模式2 “06h”：连续
测量模式3 “08h”：连续测量模式4 “
0Ah”：连续测量模式5“0Ch”：连续测
量模式6“0Eh”：连续测量模式7“10h”：
连续测量模式8“otherwise”：电源-关闭
模式

SDR位：传感器驱动设置“0”：
低噪声驱动“1”：低电源驱动

SMR位：测量范围与灵敏度设置“0”：高灵敏度
设置“1”：宽测量范围设置

STEST位：自检设置“0”：自
测试禁用“1”：自测试启用

12.3.8 BOP and BRP registers: Operating Threshold and Returning Threshold Setting of Programmable Switch Function

Addr.	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read/Write register									
22h - 25h	BOPX[7:0]	BOPX7	BOPX6	BOPX5	BOPX4	BOPX3	BOPX2	BOPX1	BOPX0
	BRPX[7:0]	BRPX7	BRPX6	BRPX5	BRX4	BRP1X3	BRPX2	BRPX1	BRPX0
	BOPY[7:0]	BOPY7	BOPY6	BOPY5	BOPY4	BOPY3	BOPY2	BOPY1	BOPY0
	BRPY[7:0]	BRPY7	BRPY6	BRPY5	BRPY4	BRPY3	BRPY2	BRPY1	BRPY0
	BOPZ[7:0]	BOPZ7	BOPZ6	BOPZ5	BOPZ4	BOPZ3	BOPZ2	BOPZ1	BOPZ0
	BRPZ[7:0]	BRPZ7	BRPZ6	BRPZ5	BRPZ4	BRPZ3	BRPZ2	BRPZ1	BRPZ0
	BOPV[7:0]	BOPV7	BOPV6	BOPV5	BOPV4	BOPV3	BOPV2	BOPV1	BOPV0
	BRPV[7:0]	BRPV7	BRPV6	BRPV5	BRPV4	BRPV3	BRPV2	BRPV1	BRPV0
Reset		0	0	0	0	0	0	0	0

Addr.	Register name	D15	D14	D13	D12	D11	D10	D9	D8
Read/Write register									
22h - 25h	BOPX[15:8]	BOPX15	BOPX14	BOPX13	BOPX12	BOPX11	BOPX10	BOPX9	BOPX8
	BRPX[15:8]	BRPX15	BRPX14	BRPX13	BRPX12	BRPX11	BRPX10	BRPX9	BRPX8
	BOPY[15:8]	BOPY15	BOPY14	BOPY13	BOPY12	BOPY11	BOPY10	BOPY9	BOPY8
	BRPY[15:8]	BRPY15	BRPY14	BRPY13	BRPY12	BRPY11	BRPY10	BRPY9	BRPY8
	BOPZ[15:8]	BOPZ15	BOPZ14	BOPZ13	BOPZ12	BOPZ11	BOPZ10	BOPZ9	BOPZ8
	BRPZ[15:8]	BRPZ15	BRPZ14	BRPZ13	BRPZ12	BRPZ11	BRPZ10	BRPZ9	BRPZ8
	BOPV[15:8]	BOPV15	BOPV14	BOPV13	BOPV12	BOPV11	BOPV10	BOPV9	BOPV8
	BRPV[15:8]	BRPV15	BRPV14	BRPV13	BRPV12	BRPV11	BRPV10	BRPV9	BRPV8
Reset		0	0	0	0	0	0	0	0

Operating threshold data of magnetic sensor X-axis/Y-axis/Z-axis
- BOPX[7:0] bits: X-axis operating threshold data lower 8-bit
- BOPX[15:8] bits: X-axis operating threshold data higher 8-bit
- BOPY[7:0] bits: Y-axis operating threshold data lower 8-bit
- BOPY[15:8] bits: Y-axis operating threshold data higher 8-bit
- BOPZ[7:0] bits: Z-axis operating threshold data lower 8-bit
- BOPZ[15:8] bits: Z-axis operating threshold data higher 8-bit
- BOPV[7:0] bits: Vector sum of 3-axis operating threshold data lower 8-bit
- BOPV[15:8] bits: Vector sum of 3-axis operating threshold data higher 8-bit

Returning threshold data of magnetic sensor X-axis/Y-axis/Z-axis
- BRPX[7:0] bits: X-axis returning threshold data lower 8-bit
- BRPX[15:8] bits: X-axis returning threshold data higher 8-bit
- BRPY[7:0] bits: Y-axis returning threshold data lower 8-bit
- BRPY[15:8] bits: Y-axis returning threshold data higher 8-bit
- BRPZ[7:0] bits: Z-axis returning threshold data lower 8-bit
- BRPZ[15:8] bits: Z-axis returning threshold data higher 8-bit
- BRPV[7:0] bits: Vector sum of 3-axis returning threshold data lower 8-bit
- BRPV[15:8] bits: Vector sum of 3-axis returning threshold data higher 8-bit

AK09973D can set Operating and Returning threshold (X, Y, Z-axis) in two’s complement. It follows the same format as Measurement data. Switch thresholds can be free to set.

12.3.8 BOP和BRP寄存器：可编程开关功能的操作阈值与返回阈值设置

地址	寄存器 name	D7	D6	D5	D4	D3	D2	D1	D0
读/写寄存器									
22h - 25h	BOPX[7:0]	BOPX7	BOPX6	BOPX5	BOPX4	BOPX3	BOPX2	BOPX1	BOPX0
	BRPX[7:0]	BRPX7	BRPX6	BRPX5	BRX4	BRP1X3 BRPX2		BRPX1	BRPX0
	BOPY[7:0]	BOPY7	BOPY6	BOPY5	BOPY4	BOPY3	BOPY2	BOPY1	BOPY0
	BRPY[7:0]	BRPY7	BRPY6	BRPY5	BRPY4	BRPY3	BRPY2	BRPY1	BRPY0
	BOPZ[7:0]	BOPZ7	BOPZ6	BOPZ5	BOPZ4	BOPZ3	BOPZ2	BOPZ1	BOPZ0
	BRPZ[7:0]	BRPZ7	BRPZ6	BRPZ5	BRPZ4	BRPZ3	BRPZ2	BRPZ1	BRPZ0
	BOPV[7:0]	BOPV7	BOPV6	BOPV5	BOPV4	BOPV3	BOPV2	BOPV1	BOPV0
	BRPV[7:0]	BRPV7	BRPV6	BRPV5	BRPV4	BRPV3	BRPV2	BRPV1	BRPV0
复位		0	0	0	0	0	0	0	0

地址	寄存器 name	D15	D14	D13	D12	D11	D10	D9	D8
读/写寄存器									
22h - 25h	BOPX[15:8]	BOPX15 BOPX14 BOPX13 BOPX12 BOPX11 BOPX10 BOPX9							BOPX8
	BRPX[15:8]	BRPX15 BRPX14 BRPX13 BRPX12 BRPX11 BRPX10 BRPX9							BRPX8
	BOPY[15:8]	BOPY15 BOPY14 BOPY13 BOPY12 BOPY11 BOPY10 BOPY9							BOPY8
	BRPY[15:8]	BRPY15 BRPY14 BRPY13 BRPY12 BRPY11 BRPY10 BRPY9							BRPY8
	BOPZ[15:8]	BOPZ15 BOPZ14 BOPZ13 BOPZ12 BOPZ11 BOPZ10 BOPZ9							BOPZ8
	BRPZ[15:8]	BRPZ15 BRPZ14 BRPZ13 BRPZ12 BRPZ11 BRPZ10 BRPZ9							BRPZ8
	BOPV[15:8]	BOPV15 BOPV14 BOPV13 BOPV12 BOPV11 BOPV10 BOPV9							BOPV8
	BRPV[15:8]	BRPV15 BRPV14 BRPV13 BRPV12 BRPV11 BRPV10 BRPV9							BRPV8
复位		0	0	0	0	0	0	0	0

磁传感器X轴/Y轴/Z轴的操作阈值数据
BOPX[7:0] 位：X轴操作阈值数据低8位 BOPX[15:8] 位：X轴操作阈值数据高8位 BOPY[7:0] 位：Y轴操作阈值数据低8位 BOPY[15:8] 位：Y轴操作阈值数据高8位 BOPZ[7:0] 位：Z轴操作阈值数据低8位 BOPZ[15:8] 位：Z轴操作阈值数据高8位 BOPV[7:0] 位：三轴矢量和操作阈值数据低

wer 8位

BOPV[15:8] 位：三轴操作阈值数据矢量和的高 8位

磁传感器X轴/Y轴/Z轴返回阈值数据 BRPX[7:0] 位：X轴返回阈值数据低8位 BRPX[15:8] 位：X轴返回阈值数据高8位 BRPY[7:0] 位：Y轴返回阈值数据低8位 BRPY[15:8] 位：Y轴返回阈值数据高8位 BRPZ[7:0] 位：Z轴返回阈值数据低8位 BRPZ[15:8] 位：Z轴返回阈值数据高8位 BRPV[7:0] 位：三轴返回阈值数据矢量和低8位 BRPV[15:8] 位：三轴返回阈值数据矢量和高8位

AK09973D 可以设置工作和返回阈值（X、Y、Z轴）为二进制补码格式。其格式与测量数据相同。切换阈值可自由设置。

Table 12.10 Vector sum of 3-axis data format

Vector sum of 3-axis [15:0] bits			Magnetic flux density [mT]	
Two's complement	Hex	Decimal	SMR bit = "0"	SMR bit = "1"
1111 1111 1111 1111	FFFF	65536	72.09	203.16
|	|	|	|	|
0000 0000 0000 0001	0001	1	0.0011	0.0031
0000 0000 0000 0000	0000	0	0	0

12.3.9 SRST[7:0]: Soft Reset

Addr.	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read/Write register									
30h	SRST[7:0]	0	0	0	0	0	0	0	SRST
Reset		0	0	0	0	0	0	0	0

SRST bit: Soft reset
"0": Normal
"1": Reset

When "1" is set, all registers are initialized. After reset, SRST bit turns to "0" automatically.

12.3.10 TEST1[15:0]/TEST2[7:0]: Test register

Addr.	Register name	D7	D6	D5	D4	D3	D2	D1	D0
Read/Write register									
40h	TEST1	-	-	-	-	-	-	-	-
41h	TEST2	-	-	-	-	-	-	-	-
Reset		0	0	0	0	0	0	0	0
Addr.	**Register name**	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
Read/Write register									
40h	TEST1	-	-	-	-	-	-	-	-
Reset		0	0	0	0	0	0	0	0

TEST1 and TEST2 register are test register for shipment test. Do not access these registers.

表 12.10 三轴数据格式的矢量和

三轴矢量和 [15:0] 位			磁通密度 [毫特斯拉]	
二进制补码	Hex	十进制	SMR位 = "0"	SMR位 = "1"
1111 1111 1111 1111	FFFF	65536	72.09	203.16
|	|	|	|	|
0000 0000 0000 0001	0001	1	0.0011	0.0031
0000 0000 0000 0000	0000	0	0	0

12.3.9 SRST[7:0]: 软复位

地址	寄存器 name	D7	D6	D5	D4	D3	D2	D1	D0
读/写寄存器									
30h	SRST[7:0]	0	0	0	0	0	0	0	SRST
复位		0	0	0	0	0	0	0	0

SRST位：软复位"
0"：正常"1"：复位

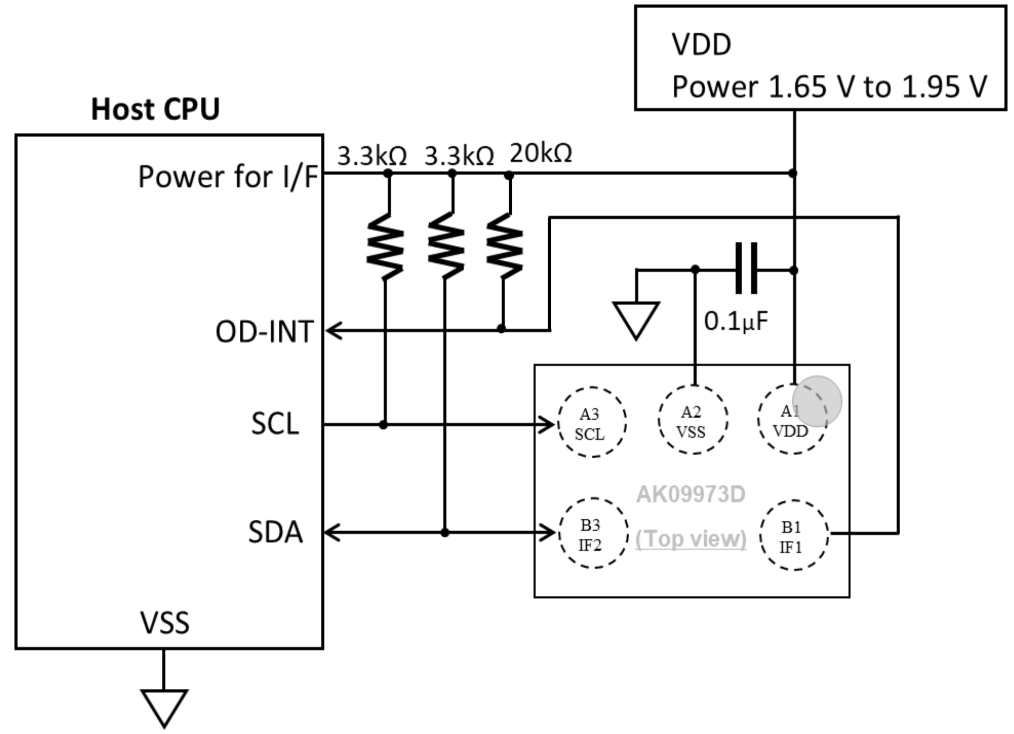
当设置为"1"时，所有寄存器 将被初始化。复位后，SRST位会自动变为"0"。

12.3.10 TEST1[15:0]/TEST2[7:0]：测试寄存器

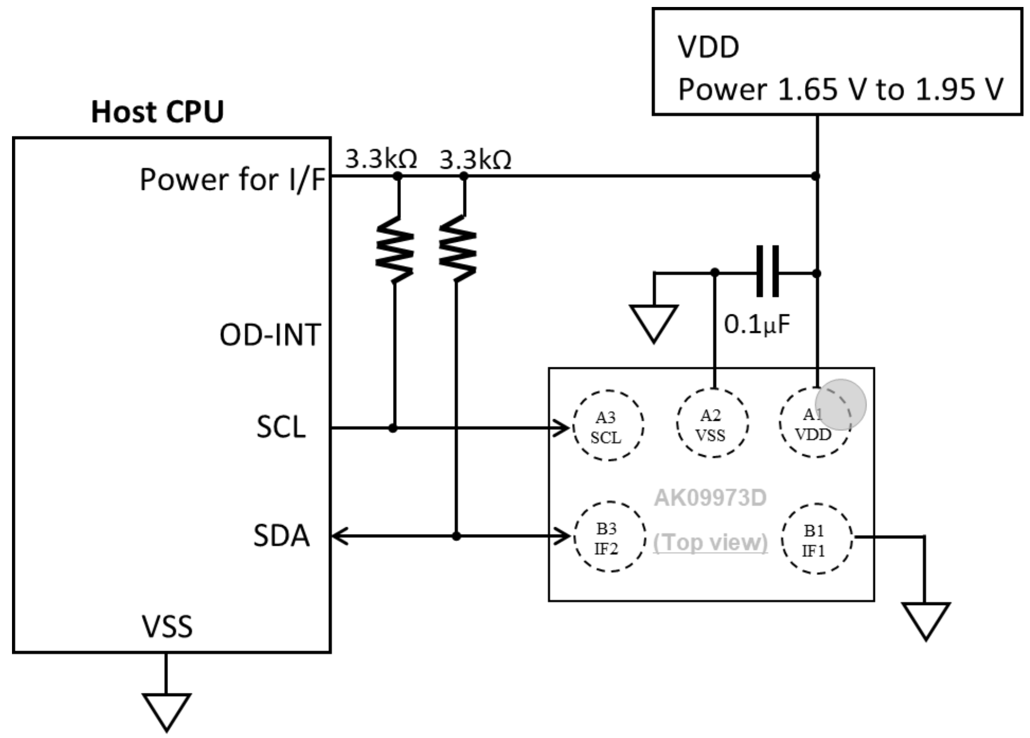
地址	寄存器 name	D7	D6	D5	D4	D3	D2	D1	D0
读/写寄存器									
40h	TEST1	-	-	-	-	-	-	-	-
41h	TEST2	-	-	-	-	-	-	-	-
复位		0	0	0	0	0	0	0	0
地址	寄存器 name	**D15**	**D14**	**D13**	**D12**	**D11**	**D10**	**D9**	**D8**
读/写寄存器									
40h	TEST1	-	-	-	-	-	-	-	-
复位		0	0	0	0	0	0	0	0

TEST1 和 TEST2 寄存器是用于出货测试的测试寄存器。禁止访问这些寄存器。 s.

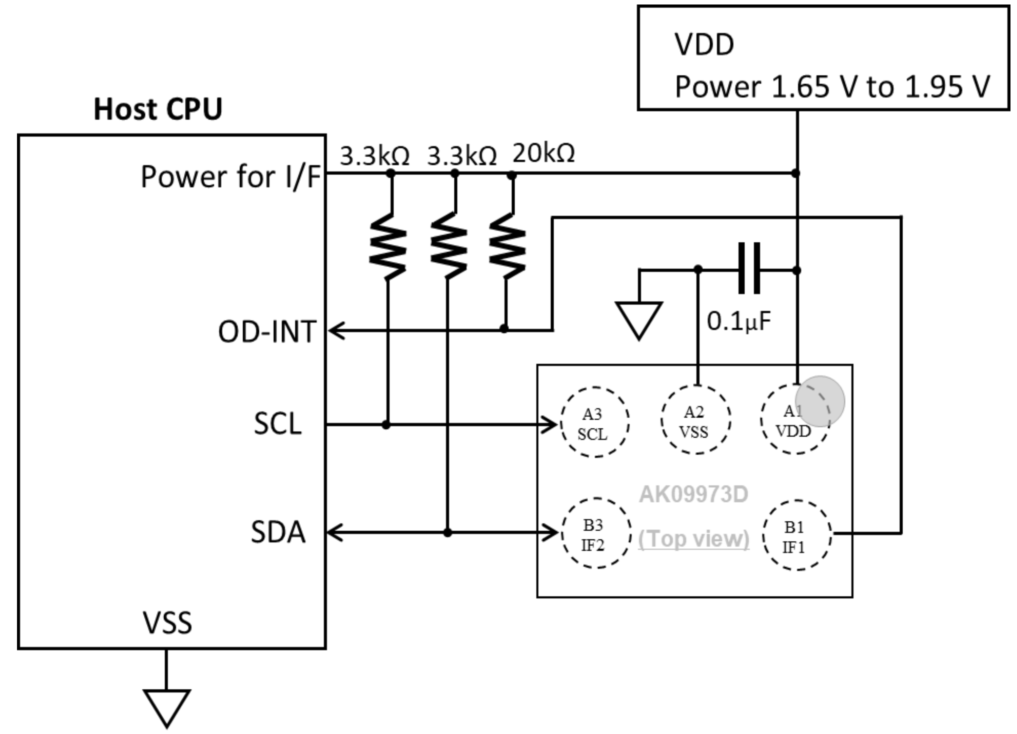
Connection1　Slave address: 10h



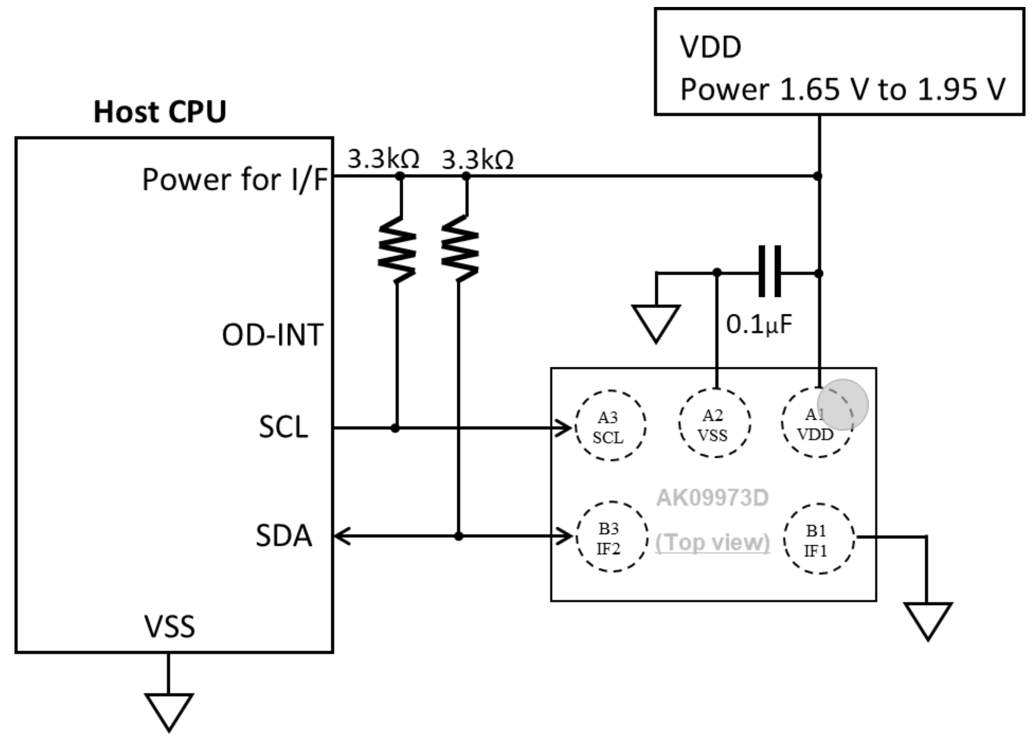
Connection1 when not using OD-INT



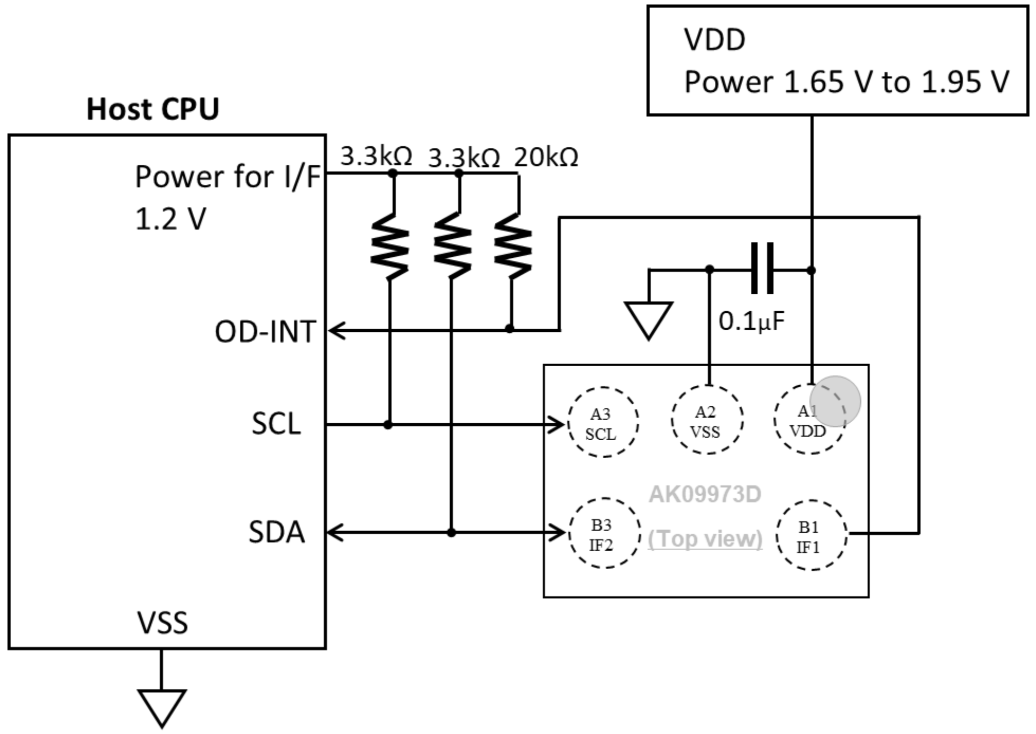
连接1 从机地址：10h



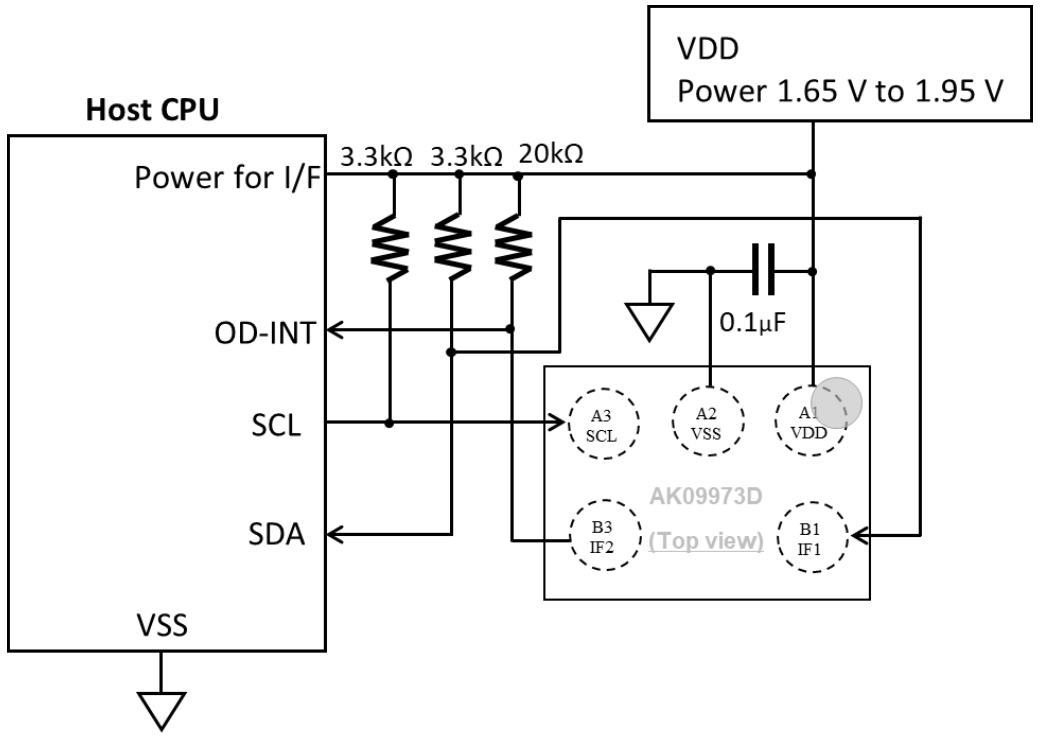
不使用OD-INT时的连接1



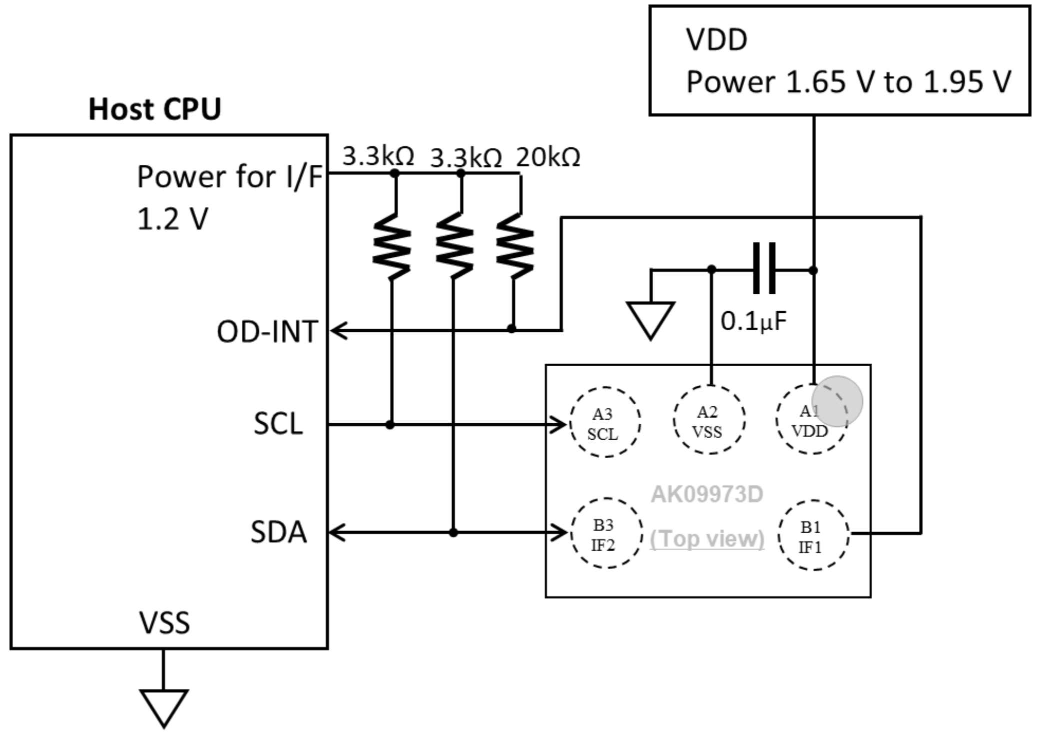
Connection1 when power for I/F is 1.2 V



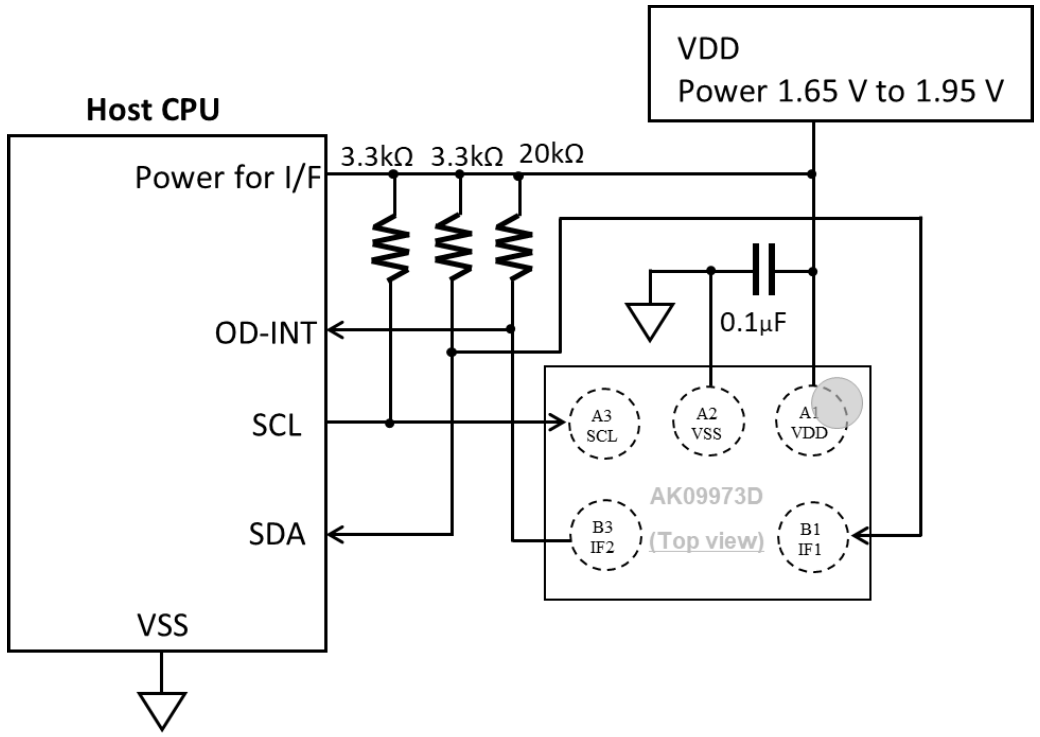
Connection2　Slave address: 11h



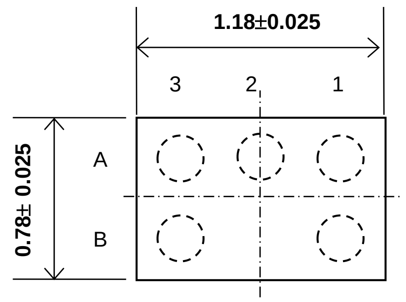
接口电源为1.2 V时的连接1



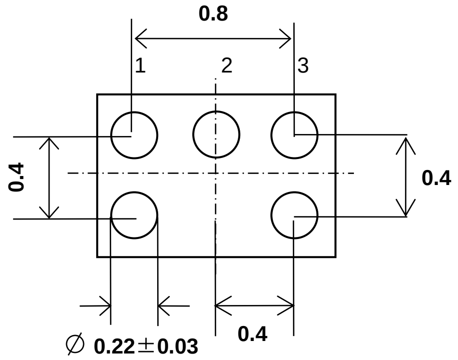
连接2 从机地址：11h



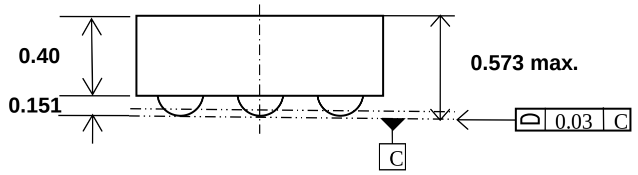
14.1. Outline Dimensions



<Top view>



<Bottom view>

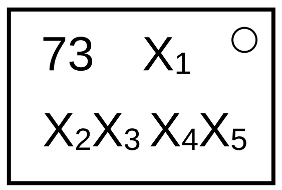


<Side view>

14.2. Marking

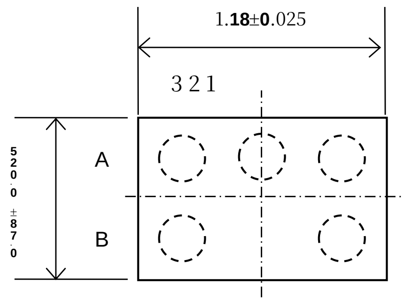
Product name: 73
Date code: $X_1X_2X_3X_4X_5$

- X1 = ID
- X2 = Year code
- X3 = Month code
- X4X5= Lot

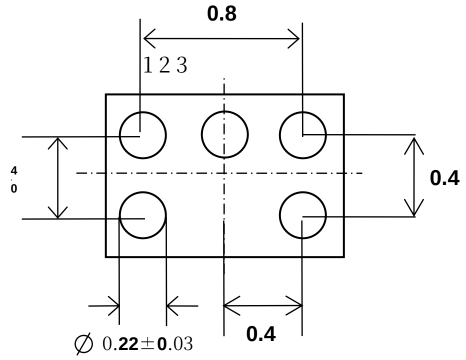


<Top view>

14.1. 外形尺寸



<顶视图>



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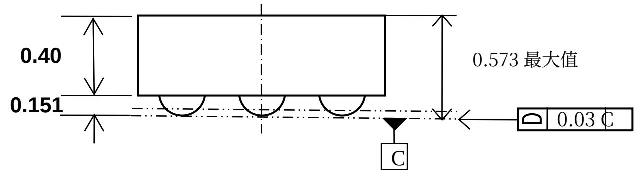
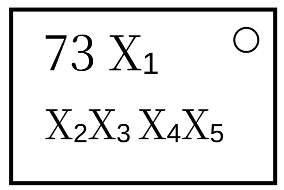


<侧视图>

14.2. 标记

产品名称： 73 日期代码：
$X_1X_2X_3X_4X_5$• X1 = 标识符 • X
2 = 年份代码• X3 = 月份
代码• X4X5= 批次



<顶视图>

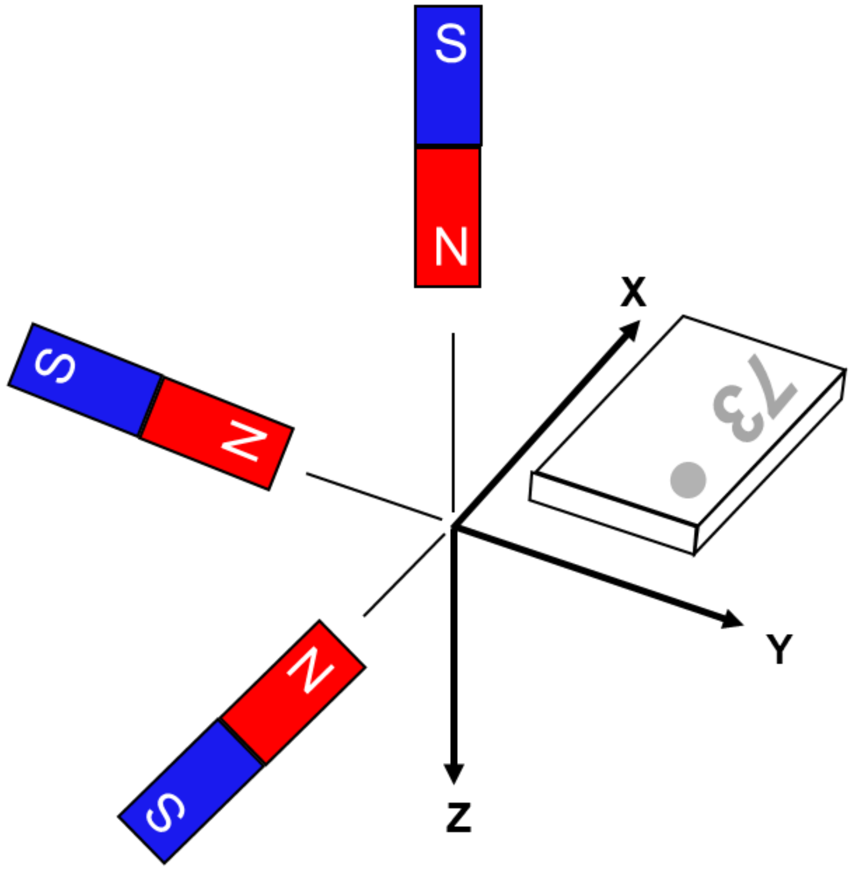
14.3. Pin Assignment

	3	2	1
A	SCL	VSS	VDD
B	IF2		IF1

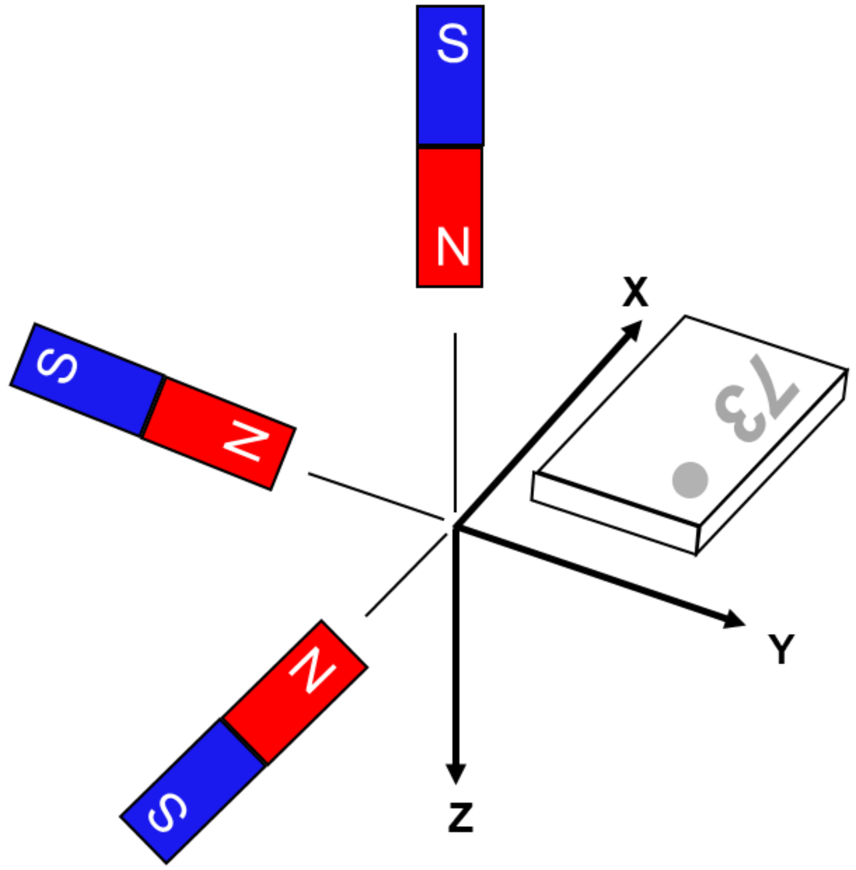
14.3. 引脚分配

	3	2	1
A	SCL	VSS	VDD
B	IF2		IF1

15. **Magnetic Orientation**



15. 磁方向



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